

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-ab96e60cc8743fbbf.jsonl`  
- SHA-256: `3b1cd5f6ead2917c002e19f3394b98d4fde8c0a2a97bcf3ab2c9a1c912dea215`  
- Source modified: `2026-06-16T04:04:01+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_47057438-9c9`  
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`
```

Transcript

1. user (2026-06-16T03:54:42.666Z)

You are a hostile red-teamer. Attack the COMBINED legal+technical recommendation below. Find every way it leaks attributable data, gives false legal comfort, breaks determinism across machines, lets a re-identification/fingerprinting attack succeed, lets minors' or persons' data slip into a public tier, or gives false confidence that a green suite proves 'no side effects'. For each attack: vector, severity, whether the design as written holds, and a concrete fix. End with an overall verdict and the must-fix-before-shipping list. Be relentless.

UNIFIED ARCHITECTURE (JSON):

```
{  
  "architectureName": "SEALED-FIXTURES: a tiered (Tier-0 public / Tier-1 key-gated), de-attributed-at-source, encrypted-where-needed regression-fixture architecture for the existing video-free eval seam",  
  "tieredModel": "TWO TIERS layered onto the EXISTING video-free seam (fusion_golden.py extracts private *_golden_features.json on the GPU box ? fusion_compare.py / ablation.py re-score them on pure CPU). The split is by SENSITIVITY, not by file format.\n\nTIER-0 ? PUBLIC, UNENCRYPTED, NO KEY. A small, de-attributed, Fork-A-only, BIOMETRIC-FREE fixture set committed under a NEW tracked dir `eval_baselines/fixtures/` (extending the existing committed-baseline precedent ? regression.py:33 already comments that eval_baselines/ is `NOT` under the gitignored output/, so a fresh checkout/CI can load a committed baseline"). ANY cloner with no key runs the SHUTTLE-ONLY regression path (experiment.predict/compare LOVO ?F1; ablation.run_ablation; the Gate-A litmus shape; regression-style floor vs GT). Contents are minimized to: opaque `fixture_id`, `fps`,`frame_width` scale constants, per-candidate `start`,`end` (time-origin reset), the 5 SHUTTLE columns (duration, peak_velocity, mean_velocity, active_ratio, net_crossings ? inanimate-object, fps/res-invariant), integer `label`, and relative `gts`. The PERSON block and AUDIO block are EXCLUDED from Tier-0 (person = the legal memo's Art.9 `tripwire` / DPDP singling-out risk in a small roster). Tier-0 tests are UNCONDITIONAL ? they run on every PR, converting the currently-skipped litmus into a real public gate.\n\nTIER-1 ? RICHER / SENSITIVE, KEY-GATED, SKIPS CLEANLY WHEN ABSENT. The full n=6+ golden corpus WITH person/audio columns (the authoritative ?F1/CI/sign-test and the exact Gate-A litmus values in test_litmus_reproduces_gate_a). Two delivery options, OUT-OF-REPO is the DEFAULT: (A) OUT-OF-REPO + TOKEN ? the *_golden_features.json stay on the existing private OneDrive store (docs/GOLDEN_DATA_SHARING.md #175); contributors/CI set `RALLY_GOLDEN_DIR` to a synced/fetched path; nothing sensitive ever enters git history (also sidesteps the legal `redistribution` act). (B) ENCRYPTED-IN-REPO (SOPS+age) only for Fork-A-clean-but-still-private data where in-repo versioning is genuinely wanted; decrypt to a gitignored path before the Tier-1 run. A Tier-1 test SKIPS (never fails) when the key/data is absent, reusing the proven `_golden_present()` skipif pattern (tests/test_ablation.py:366-371) generalized to a shared `_tier1_present()` helper. So a fresh no-key clone is always GREEN; coverage never depends on a secret being present.\n\nKEY INSIGHT (why this is elegant, not Byzantine): the two privacy properties come from DIFFERENT layers and must not be conflated. NON-ATTRIBUTABILITY comes from DELETION + RELATIVIZATION at the source (Tier-0/Tier-1 both), so it survives the analog hole ? even a fully-keyed contributor cannot re-identify, because attribution data was never written. USELESSNESS-TO-THE-NO-KEY-PUBLIC comes from minimization (opaque numbers, no schema-external meaning) PLUS optional Tier-1 encryption. Encryption does the smaller job; minimization does the
```

real work.",

"antiAttributionMeasures": [

"OPAQUE SALTED SURROGATE ID ? replace BOTH the `name` field (today = source filename via fusion_golden._derive_video_path) AND the MD5 video_id with fixture_id = first-16-hex of HMAC-SHA256(key=PER-RELEASE-SECRET-SALT, msg=video_id||name). Salt is generated once per release, kept PRIVATE (lives with the Fork-A corpus on OneDrive, NEVER in git), rotated per release. Result: (a) non-reversible without the salt, (b) NOT the MD5 video_id so holding the candidate file cannot confirm the source (kills the memo's 'video_id leak' red flag), (c) unlinkable across releases. Harness-transparent: fusion_compare.py:50 uses d['name'] ONLY as a VideoCandidates label, so aliasing it changes nothing downstream.",

"METADATA STRIP ? drop every source-attribution field: filename, video_id, match/event/tournament/player identity, capture date/clock. fps/frame_width are retained as non-identifying scale constants (shuttle features are already fps/res-invariant); quantize/bucket frame_width if counsel flags it as a small-roster/venue tell.",

"TIME-ORIGIN RESET ? for each fixture compute t0 = min over all candidate.start and gts, subtract t0 from every start/end and every gts pair (or add a random per-fixture offset). PROVABLY ZERO-COST for the current harness: metrics.py interval_iou (line 25: inter/union, both durations) and mean_start/end_error (lines 124-125: |pred-gt|) are pure differences, so a constant per-fixture offset leaves every regression metric BIT-IDENTICAL while severing the absolute match timeline that would align a fixture to a known broadcast clock.",

"STREAM MINIMIZATION (publish-by-stream) ? Tier-0 KEEPS the shuttle block + relative start/end + gts + label; EXCLUDES the entire person block (players_in_court, two_sided_motion, mean_player_motion, in_court_ratio, shuttle_player_colocation) and audio block from the PUBLIC subset. NEVER write pose/skeleton/gait/face ? already absent from the schema; the protobuf has NO field for them by design (gait survives anonymization at 80-95% re-ID per the memo).",

"RAW-TRAJECTORY EXCLUSION ? never commit the per-frame (Frame,Visibility,X,Y) trajectory CSV. Only ~5 aggregate, fps/resolution-invariant shuttle scalars per window are published ? far below the raw (x,y) stream that makes de Montjoye trajectory-linkage work (4 points ? 95% re-ID). The high-resolution fingerprint stays private.",

"PROVENANCE GATE (hard-fail) ? the de-attribution pass HARD-FAILS (refuses to emit a public fixture) on any record lacking an explicit owner-recorded (Fork-A) + minors-excluded + biometric-excluded tag verified at ingestion. A single Fork-B-tainted or minor record contaminates the public commit (memo MIXED-PROVENANCE red flag), so the gate is mechanical, not trust-based.",

"COMMITTED-LEAK REMEDIATION (do this in the SAME change) ?

eval_baselines/heuristic_lovo_n6.json currently commits real player/source names as per_video keys (adarsh_avi_singles, kushagra_singles, mahadevpura_singles, gbaaddy, ...). This is a LIVE de-attribution bug, not hypothetical. Replace those keys with opaque surrogate ids; keep the surrogate?source map PRIVATE on OneDrive.",

"PRE-COMMIT + CI LEAK GUARD ? reject committing (i) any *_golden_features.json outside the de-attributed Tier-0 allowlist, (ii) any tracked file containing a 32-hex MD5 (a possible video_id) or a known-player token or a 'video_id'/'name' source field that is not an fx_surrogate, (iii) any SOPS-encrypted file committed in plaintext (sops --verify / unencrypted-MAC check). compute_video_id (backend/utils/hashing.py) is consumed ONLY as private HMAC input and is mechanically barred from any committed file."

],

"rolloutPhases": [

"PHASE 1 (SMALL PR) ? Remediate the live leak + land the guard. Scrub eval_baselines/heuristic_lovo_n6.json (opaque surrogate keys, private mapping kept off-git); add the pre-commit + CI leak-scan (MD5/name/known-player-token detector). No new tiers yet. Self-contained, shippable, closes a real committed-attribution bug.",

"PHASE 2 (SMALL PR) ? De-attribution + minimization pass. Add scripts/deattribute_golden.py (or backend/eval/publish_fixtures.py): consumes a PRIVATE Fork-A *_golden_features.json ? emits a de-attributed, minimized, time-origin-reset *public_fixture* (opaque salted fixture_id, person/audio stripped, metadata stripped). Pure-logic, unit-testable in the dev container (verify metric-invariance of the origin reset against metrics.py). Provenance gate HARD-FAILS on un-tagged records.",

"PHASE 3 (SMALL PR) ? Tier-0 public fixtures + unconditional test. Commit a tiny de-attributed Fork-A subset (and/or a SYNTHETIC characterization set mirroring the existing _learned_corpus test builder) under eval_baselines/fixtures/; add tests/test_fixtures_tier0.py that points the EXISTING loaders at the dir and runs UNCONDITIONALLY (no skipif), folded into run_all_tests.py. Now a fresh clone has real, green, video-free coverage. (Counsel sign-off gates the REAL-subset commit; the synthetic set can land first with zero provenance risk.)",

"PHASE 4 (SMALL PR) ? Protobuf schema + deterministic serialization (OPTIONAL, only if cross-OS byte-stability or characterization snapshots are wanted). Add

backend/eval/fixtures/rally_fixture.proto (proto3, package rally.fixture.v1), the committed _gen/ module + a `make proto` + CI `protoc && git diff --exit-code` guard, a loader.py mirroring fusion_compare.load_videos, the canonical-sorted/quantized writer, .gitattributes `\*.pb binary -text`, and protobuf (Apache-2.0) in requirements. Adds the snapshot+floor replay test. Reuse the JSON path if the determinism/versioning need does not materialize ? proto is earned, not assumed.",

"PHASE 5 (SMALL PR) ? Tier-1 key-gated path. Add scripts/fetch_golden.py (OneDrive token, default) and/or a SOPS+age .sops.yaml + a `sops -d` make target; generalize _golden_present()?_tier1_present(); wire RALLY_GOLDEN_DIR through fusion_golden/fusion_compare/ablation; gate the Tier-1 CI job on the secret being present (never exposed to fork-PR runners). Tier-1 absent ? clean SKIP. Doc note in docs/GOLDEN_DATA_SHARING.md.",

"PHASE 6 (gating, NOT a code PR) ? Counsel sign-off on the final committed Tier-0 subset + the source-provenance log BEFORE the first REAL-data public commit. This is a precondition the architecture assumes, not something it satisfies."

],

"honestLimits": [

"A GREEN no-key suite PROVES only that the pure-CPU re-scoring DECISION code (gate ? optional learned filter ? merge ? IoU score) is behaviorally stable on the committed Tier-0 fixtures. It does NOT prove the GPU EXTRACTION step (video+CSV ? features in fusion_golden.py) is correct ? that is not video-free and still needs the GPU/WSL box.",

"A green Tier-0 suite does NOT reproduce the authoritative numbers. The person-cue ?F1 litmus and the exact Gate-A values (test_litmus_reproduces_gate_a) need the person-bearing Tier-1 corpus; Tier-0 publicly runs only the SHUTTLE-ONLY path. External contributors get framework-behavior + shuttle characterization coverage, not the headline fusion result.",

"This does NOT replace regression.py's DB-backed, full-pipeline-on-real-video promotion gate. It is the cheap CPU shadow that catches refactor/quality drift, not the authoritative serve-path check.",

"The data is NOT 'anonymous' in an absolute sense and the design must NEVER claim so (EDPS v. SRB / CJEU C-413/23 P). To the OWNER who can re-run the detector on the still-private source, the streams REMAIN personal data. Non-attributability here is RELATIVE (to outsiders lacking the source), CONDITIONAL (on Fork-A source staying unpublished), REVOCABLE (publish/leak the source and the link can snap back), and FORWARD-LOOKING (must be re-assessed as the public-video environment grows).",

"Encryption does NOT provide secrecy-from-a-running-contributor ? impossible by the lead's DRM/analog-hole constraint (a contributor reads plaintext + key on their machine). Tier-1 crypto is a barrier against the NO-KEY public ONLY; it must be DOCUMENTED as such and never marketed as protection it does not provide (a DPDP 'reasonable-security-safeguards'/misrepresentation question the memo flags for counsel).",

"Protobuf (or any serialization) provides ZERO secrecy ? it is an encoding, trivially decoded (protoc --decode_raw). A binary fixture format is never a confidentiality control.",

"De-attribution does NOT cure provenance or licensing. A single Fork-B-tainted or minor record contaminates the public commit regardless of how well it is de-attributed; the scheme is necessary-but-not-sufficient and sits BEHIND the per-record provenance gate + counsel sign-off, which it treats as gating preconditions, not as things it satisfies.",

"The salt is the single point of failure: the surrogate fixture_id is non-reversible ONLY while the per-release salt stays private. If the salt is ever committed/leaked alongside the public fixtures, an adversary holding candidate files can recompute the HMAC and confirm sources. Rotation per release limits blast radius to one release.",

"Time-origin reset is free for the CURRENT metrics (translation-invariant) but constrains FUTURE ones: any future scorer using absolute wall-clock, cross-fixture temporal alignment, or duration-since-match-start would break or silently mis-score on origin-reset fixtures. Flag this so an absolute-time metric is not added against public fixtures without revisiting."

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"openOwnerQuestions": [

"COUNSEL (gating): does the EU/UK sui generis database right subsist in our self-derived shuttle-coordinate/rally streams on the Football Dataco v. Sportradar 'obtained vs created' reasoning, and does redistributing a de-attributed Tier-0 subset 'extract/re-utilise a substantial part' causing CV-Online-Latvia harm? (EU AND UK.)",

"COUNSEL (gating): post Thomson Reuters v. Ross + USCO May-2025, is the US fair-use basis for the EXTRACTION step defensible given the dataset could be seen to compete in a 'data/analytics' market ? and does it matter we publish only numbers, never footage?",

"COUNSEL (gating): India ? are anti-scraping ToS enforceable and what is the IT Act s.43/s.66 exposure for any third-party-sourced ingestion; does the pending ANI v. OpenAI judgment (~April 2026) move the analysis? Confirm DPDP s.9/Rule 10 (verifiable parental consent)

and whether per-player rally-tracking is prohibited s.9(3) behavioural monitoring of a minor.",

"COUNSEL (gating): sign-off on the FINAL committed Tier-0 subset + the source-provenance log before the first REAL-data public commit; and whether non-attributability-via-private-source is a defensible posture to assert to regulators given its conditional/revocable nature.",

"OWNER (strategic): publishing the Fork-A subset FORFEITS whatever trade-secret/competitive moat it carried (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 framing, NOT US DTSA) ? weighed against the stated community/benchmark/paper-co-authorship aims. This is the owner's call.",

"OWNER (forward-looking duty): the whole posture rests on the Fork-A source staying UNPUBLISHED. Publishing a highlight reel of that same source makes TM5's correlation attack live for those records. Owner must commit to NOT publishing the source video and its derived fixtures from the same match without re-clearing, and to a periodic re-assessment as the public-video environment grows.",

"OWNER (scope): Tier-0 REAL-subset vs SYNTHETIC-only for the first public commit? Synthetic carries zero provenance risk and still exercises the full framework; a real subset adds value-level characterization but needs the full gate. Recommendation: ship synthetic first, add a counsel-cleared real subset later.",

"OWNER (mechanism): default Tier-1 delivery ? OUT-OF-REPO+token (recommended: keeps git lean, zero sensitive bytes in history, sidesteps redistribution) vs encrypted-in-repo SOPS+age (atomic versioning, offline reproducibility, heavier rotation). And: is Phase-4 protobuf wanted at all, or is canonical-JSON sufficient for the determinism/versioning need?"

],

"threatModelTable": "| # | Actor (who/what) | Capability | PROTECTED (not achievable if design holds) | NOT protected (achievable / by-design open) |\n|---|---|---|---|---|\n| TM1 | No-key public cloner | Clone repo, read committed cleartext, brute-force | Map any fixture ? source video/match/event/date/player (every attribution field stripped; id is opaque salted surrogate, not the MD5 video_id); decrypt Tier-1 (the ONE actor encryption defends); confirm a source even holding a candidate file (no MD5/timestamp/name to correlate) | Read Tier-0 shuttle scalars + relative intervals + GT and run the shuttle-only harness. By design ? Tier-0 is public. Data is useless-in-isolation (numbers with no schema-external meaning, no source pointer). |\n| TM2 | Trusted contributor WITH key, running tests | Read decrypted plaintext + key locally, re-run fusion_compare/ablation | Recover source filename, MD5 video_id, match identity, absolute timeline, face/pose/gait ? NONE are ever written into the fixture (removed at source, not gated by a key they hold). Non-attributability survives the analog hole. | Read every value the suite reads (UNPREVENTABLE per the DRM/analog-hole limit). The design does NOT pretend secrecy-from-this-actor; encryption is explicitly not represented as protecting against them. |\n| TM3 | CI runner (GitHub Actions, no human, no videos) | Decrypt if CI secret holds key; run headless; emit pass/fail + deltas | A leaked CI log/artifact cannot leak attribution it never had ? CI needs no source video, match identity, or person/biometric columns to score interval-IoU. Published subset is self-sufficient (relative intervals + shuttle + GT) so CI is green WITHOUT any stripped field. | Run the suite and read the committed fixtures (intended). Tier-1 CI secret is never exposed to fork-PR runners. |\n| TM4 | Malicious contributor (clone + key + intent; may fetch public videos + re-run our detector) | Everything TM2 can do + active correlation attacks (re-run WASB/TrackNet on candidate public videos, try to match a committed fixture) | A correlation match needs a correlatable REFERENCE. Committed subset is gated to Fork-A (owner-recorded, UNPUBLISHED), so no public reference exists ? external source-linkage 'not reasonably likely'. Even a lucky near-match carries no timestamp/name/video_id to CONFIRM, and ~5 invariant shuttle scalars/window are too coarse to back-match a 4K source. | RESIDUAL, acknowledged: protection is conditional + revocable (EDPS v. SRB / memo trap #4). Publish or leak the Fork-A source later and the link can snap back ? hence the forward-looking re-test duty. |\n| TM5 | Re-identification adversary (outside project, no key needed) | Run same deterministic detector on public videos, correlate-match a stream to its source | NOT achievable against the committed subset because: (a) source is Fork-A/unpublished (no permanent cheap public reference); (b) fixtures minimized to ~5 invariant scalars/window + relative intervals, far below the raw per-frame (x,y) that makes trajectory-linkage work ? the raw CSV is NEVER committed; (c) absolute timestamps reset to a relative origin. | Becomes LIVE if a Fork-A source is ever published ? exactly why provenance gating + the 'data-environment grows' re-assessment (NIST SP 800-188 / ICO motivated-intruder / OVIC) are load-bearing, not optional. |\n| TM6 | Accidental plaintext commit (most likely real-world failure) | A maintainer commits a decrypted *_golden_features.json or a file carrying name/video_id/match metadata | Pre-commit hook + CI scan reject any tracked file matching the private-features pattern, any 32-hex MD5 / known-player token / non-surrogate name field, and any SOPS file committed in plaintext (unencrypted-MAC check). | If the guard is bypassed/disabled, the leak lands. The guard is the mitigation; CI re-scan on every PR is the backstop. |",

"protobufSchemaSketch": "/// backend/eval/fixtures/rally_fixture.proto (proto3, package

```

rally.fixture.v1)\n// OPTIONAL layer (Phase 4) ? only if cross-OS byte-stability +
characterization snapshots are wanted.\n// Serializes EXACTLY today's *_golden_features.json
contract (fusion_golden.extract_video) plus\n// version guards, a de-attributed opaque id, and
an expected-output snapshot + quality floor.\n// NO field exists for
pose/skeleton/gait/face/bbox-track ? unrepresentable BY DESIGN.\n\nsyntax = \"proto3\";\npackage
rally.fixture.v1;\nmessage RallyFixture {\n  --- versioning / incommensurability guards
(mirror regression.py + ablation.py) ---\n  uint32 schema_version = 1;  // bump on a non-
additive change; gates the loader migration shim\n  string label_version = 2;  // Roora
guideline version (e.g. \"v8\")\n  string detector_tag = 3;  // e.g. \"wasb_hybrid@<ver>\" ?
the regression.py detector key\n  string gt_hash = 4;  // regression.gt_hash(gts),
recomputed + verified on load\n  float iou_threshold = 5;  // IoU the snapshot/floor were
computed at (default 0.5)\n\n  // --- de-attribution: NO source filename / video_id MD5 / match
/ player ---\n  string fixture_id = 6;  // OPAQUE salted surrogate (HMAC-derived hex).
NEVER the MD5 video_id.\n\n  // --- replay inputs (today's contract; time-origin reset already
applied) ---\n  float fps = 7;  // non-identifying scale constant\n  float
frame_width = 8;  // non-identifying scale constant (quantize if counsel flags)\n  repeated
Candidate candidates = 9;\n  repeated Interval gts = 10;\n\n  // --- expected outputs
for the two assertions ---\n  Snapshot expected = 11;  // blessed output of replaying this
fixture (characterization)\n  QualityFloor floor = 12;  // minimum P/R/F1 vs gts this fixture
must hold\n}\n\nmessage Candidate {\n  Interval window = 1;  // {start,end}
seconds, relative to the per-fixture origin\n  // map (not packed array) so a NEW column is
additive and an OLD fixture missing it defaults to\n  // 0.0 ? exactly
experiment._feature_column / ablation.load_videos tolerate today.\n  map<string, double> shuttle
= 2;  // duration,peak_velocity,mean_velocity,active_ratio,net_crossings\n  map<string,
double> person = 3;  // TIER-1 ONLY ? omitted entirely from the Tier-0 public subset\n
map<string, double> audio = 4;  // TIER-1 ONLY ? omitted from Tier-0\n  int32 label = 5;
// IoU-derived 0/1 (training.label_candidates), kept for parity\n  // NOTE: NO
pose/skeleton/gait/face/bbox column exists in this schema by design.\n}\n\nmessage Interval
{ double start = 1; double end = 2; }\nmessage Snapshot { string variant_spec_hash = 1;
repeated Interval segments = 2; } // sorted, quantized\nmessage QualityFloor { float
min_precision = 1; float min_recall = 2; float min_f1 = 3; }",

```

"serializationDeterminism": "Cross-OS byte-stability is enforced at four layers so a fixture and its snapshot are byte-identical on Windows/Linux/mac and CI cannot flake on float repr or CRLF: (1) FLOATS ? quantize every coordinate/feature before serialization: round(feature, 6), round(seconds, 3) (matching regression.gt_hash's round(.,3) on line 39); store the quantized values so platform float repr can never diverge. (2) FIELD/MAP ORDER ? protobuf map iteration order is unspecified, so serialize through a canonical writer that sets deterministic_serialization on the message AND sorts candidates by (start,end), feature-map keys lexicographically, and snapshot segments by (start,end). (3) NEWLINES ? fixtures are binary .pb (no newline issue); commit with `\*.pb binary -text` in .gitattributes so git never CRLF-mangles them; any sidecar text/JSON debug dump is normalized to LF. (4) ROUND-TRIP ? a test asserts serialize(parse(bytes)) == bytes for every committed fixture (canonical bytes are stable). If the project stays on JSON instead of protobuf, the SAME discipline applies via json.dump(..., sort_keys=True) + pre-quantized floats + LF ? the existing committed JSON (eval_baselines/*) already uses sort_keys=True (regression.py:98, save_baseline). The time-origin reset is itself a determinism aid: it removes large absolute timestamps whose low-order float bits are the most platform-sensitive.",

"encryptionMechanism": "Encryption is a NARROW, honestly-scoped layer used ONLY against the no-key public (TM1) ? it is belt, not parachute; the upstream minimization does the real work. RECOMMENDATION: SOPS + age for any encrypted-in-repo Tier-1 secrets (per-recipient public-key encryption, git-diff-friendly value-level for JSON, trivial CI/contributor key handling, an explicit decrypt step so there is no silent-plaintext-commit failure mode, and a MAC that detects tampering). age over GPG for ergonomics (no keyring/agent ceremony). REJECTED alternatives: git-crypt (one shared symmetric key ? painful rotation, opaque binary diffs, a misconfigured checkout silently commits plaintext); git-lfs+encryption (LFS solves large-binary storage not secrecy ? our feature JSONs are KB-scale, so it buys nothing and reinvents git-crypt with a server dependency). PREFERRED OVERALL for Tier-1: OUT-OF-REPO + token (the existing OneDrive store) ? keeps ZERO sensitive bytes in immutable git history (a future key compromise can't decrypt old commits, and the legal 'redistribution into a public repo' act never happens for sensitive data) and reuses the #175 RALLY_GOLDEN_DIR seam. Pick per-fixture: Fork-A-clean-and-stable ? in-repo SOPS is fine; anything Fork-B-tainted or identity-adjacent ? out-of-repo only. CRYPTO LIBRARY note: the surrogate-id HMAC-SHA256 uses Python's stdlib hmac/hashlib (no new dependency, license-clean); SOPS+age are external CLI tools invoked from a make target, not Python deps. DOCUMENT EXPLICITLY in code comments + docs/GOLDEN_DATA_SHARING.md that Tier-1 crypto defends ONLY against the no-key public and that a running contributor necessarily holds

plaintext+key ? stating this plainly heads off the DPDP 'reasonable security safeguards'/misrepresentation question the memo flags.",

"keyManagement": "TWO secrets, by Tier-1 delivery mode. (A) OUT-OF-REPO+token (default): a scoped OneDrive read token. CI = repository secret RALLY_GOLDEN_TOKEN, exposed only to the Tier-1 job on trusted branches, NEVER to fork-PR runners (a malicious fork PR cannot exfiltrate it). Contributor onboarding = granted scoped read access to the private store. Rotation = revoke the share / rotate the token. (B) Encrypted-in-repo (SOPS+age): an age keypair per recipient. CI = the age PRIVATE key as repository secret RALLY_AGE_KEY, Tier-1 job only, never to fork runners. Contributor onboarding = the contributor runs age-keygen and sends their PUBLIC key; a maintainer adds it to .sops.yaml and runs `sops updatekeys` ? NO existing secret is reshared, each recipient decrypts with their own private key. Rotation = edit .sops.yaml recipients (remove a departed/compromised key, add the new) + `sops updatekeys`; because age is per-recipient, removing a recipient is a one-command re-key (vs git-crypt's whole-repo re-encrypt where the old key still decrypts history). THE SALT (separate from the encryption key, and more important): the per-release HMAC salt that makes fixture_id non-reversible lives with the PRIVATE Fork-A corpus on OneDrive, NEVER in git, rotated per public release so a single salt leak does not retro-link all releases. On a suspected DATA leak: rotate the token/re-key AND ? because non-attributability is revocable ? re-assess whether the underlying Fork-A source must stay unpublished. A PRIVATE surrogate?source map is kept in the out-of-repo store for owner/CI debugging and is NEVER committed.",

"ciIntegration": "TWO CI jobs with a hard split so coverage never depends on a secret: (1) TIER-0 job ? runs on EVERY PR including fork PRs, no key, executes tests/test_fixtures_tier0.py + the unconditional fixture-replay test against eval_baselines/fixtures/ (folds into the existing run_all_tests.py / pytest, no new runner). This is the public always-green gate. (2) TIER-1 job ? gated on the secret being present (RALLY_GOLDEN_TOKEN or RALLY_AGE_KEY), runs only on trusted branches (never fork-PR runners), fetches/decrypts the corpus to a gitignored path, runs the full ?F1/CI/sign-test + the exact Gate-A litmus. When the secret is absent the Tier-1 tests SKIP cleanly via _tier1_present() (generalizing tests/test_ablation.py:366-371). GUARD JOBS on every PR: (a) the leak scan ? reject any committed 32-hex MD5 / known-player token / non-surrogate name field / private-features filename / plaintext-committed SOPS file; (b) if protobuf is adopted, re-run `protoc` and `git diff --exit-code` to prove the checked-in _gen/ module matches the .proto. All of this is offline, pure-CPU, GPU-free ? runs in the dev container (the project's full pipeline cannot, but this layer can).",

"contributorWorkflow": "END-TO-END, contributor on another clone with NO video present: (1) git clone ? gets the public repo including eval_baselines/fixtures/ (Tier-0). (2) pip install -r requirements.txt (+ protobuf if Phase 4 landed) ? pure-CPU deps, no GPU/torch/cv2 needed for this path. (3) python run_all_tests.py ? the UNCONDITIONAL Tier-0 tests run immediately: the existing loaders (fusion_compare.load_videos / ablation.load_videos, schema-compatible) read the committed fixtures, replay them through the REAL experiment.predict/compare/run_ablation + metrics.score code, and assert the shuttle-only LOVO ?F1 framework behavior (and, if Phase 4, a byte-stable snapshot + quality floor). GREEN with no key, no video, no DB. (4) The Tier-1 tests SKIP cleanly with a clear reason ('Tier-1 golden corpus absent: set RALLY_GOLDEN_DIR or provide the key'). (5) A TRUSTED contributor additionally: is granted scoped OneDrive read access (or sends an age public key for SOPS), runs scripts/fetch_golden.py (or `sops -d` / `make decrypt-golden`) to populate RALLY_GOLDEN_DIR, then re-runs ? now the full person/audio ?F1, CI, sign test, and the exact Gate-A litmus values execute. At NO point can any contributor ? keyed or not, trusted or malicious ? recover a source video, match, date, or player from the fixtures, because that data was deleted at the source, not encrypted. They CAN read every number the tests read (the analog-hole limit, accepted by design).",

"harnessIntegration": "Purely ADDITIVE on existing seams ? no registry fork, no schema fork, no change to experiment.py/regression.py/ablation.py/fusion_compare.py internals. MODULES TO EXTEND / ADD: (1) backend/eval/fusion_compare.py + backend/eval/ablation.py ? already read ONLY c['shuttle'][...]/c['start']/c['end']/d['gts'] and use d['name'] as an opaque label (verified fusion_compare.py:35-52, ablation.py:386-407); the Tier-0 public loader iterates only SHUTTLE columns (skips PERSON/AUDIO), a small additive code path; the existing skipif litmus tests keep running against the PRIVATE person-bearing Tier-1 features unchanged. (2) eval_baselines/ ? the committed-structured-data precedent (regression.py:33); add the tracked subdir eval_baselines/fixtures/ alongside regression_baseline.json / experiment_leaderboard.json / heuristic_lovo_n6.json. (3) backend/eval/regression.py ? reuse gt_hash (line 37) for the fixture's gt_hash guard and its IoU/detector incommensurability checks (lines 132-141); reuse the committed-baseline floor pattern. (4) backend/eval/experiment.py ? reuse Variant.spec_hash() (line 148) as the snapshot's variant key and _feature_column's 0.0 default (line 180) for the additive-column tolerance; the replay calls predict/compare/run_ablation UNCHANGED. (5) backend/eval/metrics.py ? the translation-invariance that makes the time-origin reset free is a property of score()'s pure-difference math (lines 25,124-125); no change needed, just relied

upon. (6) backend/utils/hashing.py::compute_video_id ? the canonical MD5 source, consumed ONLY as private HMAC input, mechanically barred from any committed file. NEW FILES:
 scripts/deattribute_golden.py (the de-attribution+minimization pass),
 tests/test_fixtures_tier0.py (unconditional), the leak-scan pre-commit hook; OPTIONAL Phase-4:
 backend/eval/fixtures/{rally_fixture.proto, loader.py, _gen/rally_fixture_pb2.py},
 scripts/build_fixtures.py, tests/test_fixture_replay.py, .gitattributes, +protobuf in
 requirements; OPTIONAL Phase-5: scripts/fetch_golden.py + .sops.yaml. COMPOSES with the
 concurrent feat/tier0-rally-seg-eval worktree (currently at master HEAD, no fixtures landed):
 that workflow owns the runnable-without-video MECHANICS; THIS layer is the LEGAL/privacy FILTER
 deciding which fields are publishable and under what id ? mechanics produce the runnable
 fixture, this de-attributes + minimizes it before the public commit."
 }

LEGAL MEMO (JSON):

```
{
  "executiveSummary": "INTERNAL MEMO ? general legal information, NOT legal advice; retained
India + EU/UK counsel sign-off required before any public commit. QUESTION: can the project
commit DERIVED numeric data (per-frame shuttle (x,y) coordinates, rally start/end time-
intervals, audio/person/court feature columns, ground-truth intervals) into a public GitHub repo
so external contributors can run the rally-segmentation regression suite WITHOUT the videos?
BOTTOM LINE: Yes, but only under conditions, and the analysis FORKS on data provenance. The
OUTPUT data itself is, across all three jurisdictions, essentially uncopyrightable
FACT/measurement ? it does not inherit the source video's copyright (US: Feist + 17 USC 102(b) +
NBA v. Motorola; India: EBC v. D.B. Modak, no sui generis right; EU: Football Dataco v. Yahoo ?
copyright protects only original selection/arrangement, never the numbers). So a thin
compilation copyright at most attaches to a creative schema, never to the coordinates. BUT
copyright is the easy axis and NOT where the risk lives. The real exposure is fourfold and
provenance- and stream-dependent: (1) the ACT of ingesting/extracting from the source video
needs its own lawful basis ? owned/permissioned video is clean; a US fair-use defense for
analytic extraction is now CONTESTED, not \"strongly favorable,\" after Thomson Reuters v. Ross
(2025) and the US Copyright Office May-2025 report, and unlawful acquisition independently
destroys the defense (Bartz v. Anthropic, ~$1.5B settlement); (2) CONTRACT / Terms-of-Service
can bar extraction and redistribution even where copyright cannot reach the facts (ProCD; hiQ v.
LinkedIn; YouTube ToS bars scraping AND \"harvest information that might identify a person (for
example... faces)\") AND redistribution) ? and the remedy is injunction + forced deletion of the
committed dataset, not nominal damages; (3) EU/UK sui generis DATABASE RIGHT may subsist in
self-derived match-event/coordinate streams because, on the leading Football Dataco v.
Sportradar reasoning, recording pre-existing on-court facts you do not control is \"OBTAINING,\"
not \"creating\" ? so the project's comfortable \"we created it\" premise is genuinely
contestable; (4) PRIVACY ? person/pose/gait streams and anything involving MINORS are the
highest-risk and several should NOT be published at all. The cleanest defensible path: commit
ONLY owned-or-permissioned-source, minors-excluded, person/pose/gait-EXCLUDED (shuttle +
abstract rally-interval + non-biometric feature) data, stripped of every source-attribution
field (no filename, no video_id MD5, no match metadata), under a no-extraction-from-third-party-
source and no-redistribution hygiene posture, with optional encryption only as a barrier against
the no-key public (it cannot bind a contributor who runs the tests).\",
  \"derivedDataCoreVerdict\": \"The derived numeric data DOES NOT inherit the source video's
copyright in any of the three jurisdictions ? this is the one firmly-settled conclusion. Per-
frame shuttle (x,y) coordinates, rally start/end timestamps, and audio/person/court feature
values are uncopyrightable FACTS/measurements about a recorded physical event: discovered, not
authored. US: Feist (facts never copyrightable; \"originality, not sweat of the brow, is the
touchstone\"), 17 USC 102(b) (no copyright in any \"discovery\"), NBA v. Motorola (facts
generated by a sporting event are uncopyrightable \"purely factual information which any
patron... could acquire from the arena\" even though the broadcast is protected). A
coordinate/timestamp table is also NOT a 17 USC 101 \"derivative work\": a derivative work
recasts the original's protected EXPRESSION (framing, editing, imagery, commentary), and a
number table carries none of it. India: EBC v. D.B. Modak ? copyright needs \"skill and
judgment,\" protects only original arrangement, not facts; no EU-style sui generis right exists.
EU: Football Dataco v. Yahoo (C-604/10) ? database copyright protects only an original
selection/arrangement; \"the intellectual effort and skill of creating that data are not
relevant\" and sweat-of-the-brow cannot ground it. CEILING ON PROTECTION: the MOST that can
attach is a THIN compilation copyright in a genuinely creative SELECTION/ARRANGEMENT (e.g., a
curated schema) ? never in the raw numbers, and a mechanical/exhaustive layout (every frame,
every rally) likely has too little originality even for that (Feist's \"garden-variety listing\"
problem). CRITICAL SEPARATION: the STATUS of the output data (settled: unprotected facts) is a
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different question from the ACT of extraction (transiently copying the video to compute the data), which can independently infringe (Sega v. Accolade: intermediate copying can infringe even when the end product reproduces no protected expression) and which needs its own license or fair-use basis. NON-COPYRIGHT FENCES survive precisely BECAUSE the data is uncopyrightable: enforceable no-extraction ToS (ProCD), narrow hot-news misappropriation (NBA v. Motorola), and the EU/UK sui generis database right. So \"the numbers are free facts\" answers the copyright question and ONLY the copyright question.",

```
"byJurisdiction": [
  {
    "jurisdiction": "India (PRIMARY ? owner is Bangalore-based, commercialization intended)",
    "copyright": "Derived facts are NOT copyrightable as such: EBC v. D.B. Modak (2008 SC) requires 'skill and judgment' and protects only original arrangement, not facts; India has NO EU-style sui generis database right. BUT this is not as clean as it looks: Indian High Courts protect DATABASES/COMPILATIONS as 'original literary works' under s.2(o) on a low 'skill, labour and judgment' threshold ? Burlington Home Shopping v. Rajnish Chibber, V. Govindan v. E.M. Gopalakrishna Kone, and the OLX v. Padawan order itself (which held a scraped dataset to be a protectable 'proprietary database'). So if numbers are LIFTED from a third party's structured dataset/feed, copying its selection/arrangement can infringe even though the raw facts are free. Self-derived-from-pixels data avoids this; copying an official stats compilation does not.",
    "databaseRight": "No sui generis database right exists in India ? a genuine permissiveness advantage over the EU/UK. Protection runs only through copyright (compilation, above) or contract/IT-Act.",
    "contractToS": "The LIVE risk, not copyright. Breach of anti-scraping ToS plus IT Act s.43 (Government has affirmatively argued public-data scraping 'may be in breach of section 43') and s.66 layering CRIMINAL exposure (up to 3 years / Rs 5 lakh). Enforceability of anti-scraping ToS is untested in Indian courts. OLX v. Padawan was decided AGAINST the scraper; its 'private use' line is obiter, not a safe harbor. Trade-secret/confidentiality for the owner's OWN corpus runs NOT through any US-style DTSA (India has no trade-secret statute) but through the Indian Contract Act 1872, equity, common-law breach of confidence, and IT Act s.72 ? and publication still forfeits the moat under that doctrine.",
    "privacy": "DPDP Act 2023 governs personal data of identifiable natural persons; genuinely anonymized numbers fall outside it, but the s.3(c)(ii) 'publicly available data' carve-out is CONTESTED (MeitY/Minister has said scrapers must still meet DPDPA consent obligations). MINORS are stricter than the EU: 'child' = under 18 (s.2(f)); s.9(1) requires VERIFIABLE parental consent (DPDP Rules 2025 Rule 10 ? real ID/age verification, not a checkbox); s.9(2) bars detrimental processing; s.9(3) PROHIBITS tracking/behavioural monitoring/targeted ads of children ? subject to narrow Rule 12/Fourth-Schedule class/purpose exemptions that do NOT cover an amateur highlight indexer, so the ban DOES bite this project. A persistent per-player pose/gait signature is, on the conservative reading, prohibited behavioural monitoring; 'tracking' is undefined in the Act, so this is the defensible-conservative, not the certain, reading.",
    "netVerdict": "MORE PERMISSIVE than the EU on the database-right axis, but NOT 'lawful on its face.' Defensible to publish ONLY genuinely-anonymized, independently-derived (not lifted from a third-party compilation) numbers, with NO minors tracked and accepting that anti-scraping-ToS + IT Act s.43/s.66 exposure for any third-party-sourced ingestion is unresolved. Owned-source + minors-excluded + no-person/pose/gait = clean."
  },
  {
    "jurisdiction": "United States (repo + product are global; US fair-use & contract law are the most-developed)",
    "copyright": "Output data does not inherit the video's copyright (Feist; 102(b); NBA v. Motorola); not a 101 derivative work; thin compilation copyright at most. The EXTRACTION act is the open question: fair use for analytic, non-expressive copying was historically favorable (Authors Guild v. Google; Google v. Oracle) but is now CONTESTED ? Thomson Reuters v. Ross (D. Del. 2025) REJECTED the 'copy an expressive work just to reach unprotected facts/data' theory, distinguished the functional-CODE cases (Sega/Oracle), found the use non-transformative, and found Factor-4 harm in a derivative 'data to train AIs' market the plaintiff had not even entered; the US Copyright Office (May 2025) rejected 'non-expressive therefore inherently transformative' and backs a market-dilution harm theory. If the dataset competes with a rights-holder's own data/analytics market, fair use may FAIL. Unlawful source acquisition independently defeats the defense regardless of how transformative the end use is (Bartz v. Anthropic, 2025; ~$1.5B settlement).",
    "databaseRight": "No sui generis database right in the US (Feist rejected sweat-of-the-brow). Competitors stay free to copy raw facts ? subject to the non-copyright fences below.",
    "contractToS": "The dominant US risk. Post-Van Buren the CFAA no longer reaches mere
```

purpose-violations of accessible/public data (though footnote 8 RESERVED whether contract/policy limits can still ground CFAA liability ? the federal-crime route is narrowed, not dead), so platforms enforce via CONTRACT. ToS is an enforceable contract that bites where copyright/CFAA fail (ProCD; hiQ v. LinkedIn ? breach-of-contract summary judgment + \$500k consent judgment + injunction + ORDER TO DESTROY scraped-derived data). Binding turns on assent: a logged-in/account/API-key or genuine clickwrap user is bound (hiQ; any YouTube-API user); a purely logged-out scraper under browserwrap-only may not be (Meta v. Bright Data) ? and thin 'sign-in wrap' flows can fail for want of conspicuous notice. YouTube ToS verbatim bars automated access, 'harvest any information that might identify a person (for example... faces)', and redistribution/commercial reuse. REDISTRIBUTION (committing the dataset publicly) is the single highest-risk act and the remedy is injunction + forced deletion, not nominal damages.",

"privacy": "GDPR-style analysis below applies to global users; US state biometric statutes (e.g., Illinois BIPA ? not separately briefed here) and publicity/personality rights are additional fences for face crops and identifiable-player gait. Faces in any committed imagery = identifying personal data; do not publish.",

"netVerdict": "Output data publishable as facts ONLY if (i) source video was owned/permissioned (or extraction has a defensible fair-use basis you do NOT over-rely on), (ii) no enforceable ToS bars extraction/redistribution of the ingested source, and (iii) no person/pose/gait/face/minor stream is included. Treat third-party-ToS-sourced derived data as NOT publicly committable."

},

{

"jurisdiction": "EU / UK (product/repo global; English-language highlight market makes UK a likely target)",

"copyright": "Database copyright protects only an original selection/arrangement, never the numbers, and never sweat-of-the-brow (Football Dataco v. Yahoo, C-604/10). Self-derived coordinates are facts, not a copied original structure ? so copyright is a low axis.",

"databaseRight": "The genuinely contestable axis. The sui generis right protects investment in OBTAINING/verifying/presenting pre-existing data, NOT in CREATING it (BHB v. William Hill C-203/02; Fixtures Marketing) ? which the project reads as 'we create coordinates frame-by-frame, so we are clear.' THAT READING IS OVERCONFIDENT: in Football Dataco v. Sportradar the live match-event data (goals/scorers) was held to be OBTAINED, because the recorder captured pre-existing facts ('events taking place on the pitch') it did NOT control. A badminton rally/shuttle position is the same kind of pre-existing on-court fact recorded rather than authored, so a rights-holder can argue the right SUBSISTS in our streams. Infringement is now governed by CV-Online Latvia (C-762/19, 2021): even a substantial-part taking is actionable only where it adversely affects the maker's ability to recoup investment, balanced against information flow/competition ? a fact-intensive harm test, not a clean 'no protected part = safe' rule. Sportradar (C-173/11) also defeats the server-location dodge for EU-targeted re-utilisation. UK retained its own sui generis right and pre-2021 CJEU case law, so UK-targeted products carry AT LEAST equal, arguably greater, risk; a non-UK/EEA maker may not even hold the right defensively, which does nothing to reduce exposure as a potential infringer.",

"contractToS": "Ryanair v. PR Aviation (C-30/14): where a database enjoys NEITHER copyright NOR the sui generis right, the Directive's user-protections (incl. no-contracting-out) do not apply, so the source's T&Cs/licence can lawfully BAN automated extraction and redistribution and bind anyone who agreed ? the dominant residual risk. Trap: the licence that legitimises INGESTION of the footage is often the very instrument that contractually bans extraction/redistribution of derived data.",

"privacy": "GDPR: shuttle/object coordinates are NOT personal data in isolation (Art.4(1)) ? but coordinates published as part of an identifiable named match are part of a personal-data operation. Person bounding boxes that single out a human are ordinary personal data, NOT Art.9 biometric UNLESS used to uniquely identify (EDPB Guidelines 3/2019). Pose/gait is the danger zone: gait re-identifies even with faces blurred and skeleton 'anonymization' leaves gait identity intact (Weiss et al., Sensors 2025) ? Art.9 special-category the moment used to uniquely identify; explicit-consent basis required. 'Anonymous' is RELATIVE post EDPS v. SRB (CJEU C-413/23 P, 4 Sept 2025): a stream can be non-attributable to a recipient lacking the source yet remains PERSONAL DATA to the holder who can re-run the detector. Minors: Art.8 (consent floor 13?16); any biometric/special-category processing of a child stacks Art.9.",

"netVerdict": "Do NOT assume self-derivation defeats the database right ? treat self-derived event/coordinate streams as potentially 'obtained' data and avoid taking in a way that could reconstitute a rights-holder's dataset or harm their feed's monetisation. Read every source's T&Cs for extraction/redistribution clauses. Keep the footage-copyright analysis (highlight reels) strictly separate. Person/pose/gait/face = do not publish without an Art.9 explicit-consent basis. Get EU + UK jurisdiction-specific advice before redistributing numeric streams at scale."

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    }
  ],
  "provenanceFork": "PROVENANCE IS THE DECISIVE FORK ? the legal posture of an identical numeric
file flips entirely on where the source video came from. FORK A ? OWNED / OWNER-RECORDED (or
fully-permissioned/licensed-for-this-use) SOURCE: This is the clean path and the only one safe
to commit publicly. No third-party copyright in the source bars the extraction; no third-party
ToS governs the owner's own recording (with two caveats below); the EU/UK database-right
'obtained vs created' debate still applies in theory but there is no adverse rights-holder to
assert it against the owner's own footage. The analysis collapses to: (1) a TRADE-SECRET /
competitive-moat CHOICE (US framing: 18 USC 1839 DTSA requires secrecy measures + value-from-
secrecy; the OPERATIVE framing for this Bangalore owner is Indian breach-of-confidence /
Contract Act 1872 / IT Act s.72, NOT the DTSA) ? publishing the data forfeits whatever moat it
carried, which is the owner's strategic call to make, weighed against the stated
community/benchmark/paper-co-authorship aims; and (2) PRIVACY ? the owner's own footage still
contains identifiable people, so DPDP/GDPR and the minors rules below still bind, and
person/pose/gait/face streams are still off-limits. TWO CAVEATS that narrow Fork A: (a) 'owner-
recorded' is clean ONLY for the owner's OWN non-ticketed/private sessions; footage the owner
SHOT AT a third party's event (tournament, club match) carries live exposure ? a ticket is a
contract of admission that routinely bars recording/redistribution, and
venue/organiser/broadcaster rights (e.g., BWF/league terms, Rome Convention rebroadcast rights)
attach independently of any platform ToS; (b) any third-party-collected footage in the corpus
(the project's golden set may mix sources) taints those records and pushes them into Fork B.
FORK B ? THIRD-PARTY-SOURCED (broadcast / YouTube / scraped / downloaded) SOURCE: Derived data
from this provenance SHOULD NOT be committed to a public repo. Even though the numbers are
uncopyrightable facts, (1) the EXTRACTION act needs a license or a now-CONTESTED US fair-use
defense, and unlawful acquisition (e.g., scraping against ToS, downloading from a platform that
bars it) independently sinks the defense (Bartz v. Anthropic); (2) the source's ToS (YouTube et
al.) typically bars automated extraction AND redistribution, and committing the derived dataset
publicly IS redistribution ? the highest-risk act, remediable by injunction + forced deletion of
the committed data (hiQ); (3) re-utilisation/extraction of a substantial part may infringe the
EU/UK sui generis database right if the source ties back to a protected feed. PRACTICAL RULE FOR
THE REPO: gate the committed corpus to FORK A only. Tag every golden-set record with a verified
provenance flag at ingestion; only owned/permissioned-source, non-event-ticketed, minors-
excluded, person/pose/gait-excluded records are eligible for the public commit. This maps
directly onto the existing split: eval_baselines/ (aggregate JSON snapshots) is already
committed precedent and is provenance-agnostic if it carries no per-source attribution;
*_golden_features.json are currently kept private (docs/GOLDEN_DATA_SHARING.md) precisely
because they derive from a 'Proprietary corpus' ? the right move is to keep Fork-B-tainted
features private and publish only a Fork-A, de-attributed subset.",
  "privacyVerdict": "PUBLISH/NO-PUBLISH BY STREAM (the project's guardrails already point the
right way; this hardens them). PUBLISH-OK (lowest risk): (1) SHUTTLE (x,y) coordinates +
Visibility ? describe an inanimate object, NOT personal data on their own (GDPR Art.4(1); WP136
content/purpose/result test) ? PROVIDED they are NOT joined to anything that singles out a named
player and NOT published as an identifiable named match (the surrounding personal-data
obligations survive even when the coordinate column alone is object-data). (2) Abstract RALLY
start/end time-INTERVALS and ground-truth intervals stripped of match/player identity ? facts
about event timing. (3) Non-biometric AUDIO-ONSET, COURT-BOUNDS, and aggregate PERSON
COUNT/feature columns used ONLY for rally detection and DETACHED from any identity ? ordinary
low-risk data with normal hygiene; NOT Art.9 because the purpose is detect/categorize, not
uniquely identify (EDPB Guidelines 3/2019). CONDITIONAL ? only with an Art.9 explicit-consent
basis (and effectively never for this v1): PERSON bounding-box coordinates that persistently
single out and follow a specific player across rallies ? the purpose flips to unique
identification and Art.9 attaches; this architecture is PRONE to that flip, so treat per-player
tracking as a tripwire. DO NOT PUBLISH (high risk, exclude from any committed stream): (4) POSE
keypoints / SKELETON / GAIT signatures ? gait is a behavioural biometric that re-identifies
people even with faces blurred, and pose 'anonymization' provably leaves gait identity intact
(Weiss et al., Sensors 2025, 80?95% gait-recognition on anonymized data); raw pose/gait tracks
are NOT reliably anonymous, are personal data when usable to single out a player, and become
Art.9 special-category once used to uniquely identify ? this matches the project's existing
'gait/biometrics strictly FUTURE (GDPR Art.9)' guardrail. (5) Any FACE CROP or identifiable
IMAGERY ? identifying personal data under GDPR and DPDP, and barred by YouTube ToS's face-
harvesting clause; do not commit imagery at all. (6) Anything that joins a NAMED PLAYER to a
gait/pose signature. MINORS FLIP EVERYTHING: under DPDP a 'child' is anyone under 18; processing
ANY child's data needs VERIFIABLE parental consent (Rule 10 ? real ID/age check, not a
checkbox), must not cause detrimental effect, and s.9(3) PROHIBITS behavioural

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monitoring/tracking of children (Rule 12/Fourth-Schedule exemptions do not cover this project, so the ban applies). A persistent per-player pose/gait signature is, conservatively, prohibited monitoring. CLEANEST PATH for any footage containing minors: EXCLUDE minors entirely from any persisted derived stream beyond inanimate shuttle/interval data ? consistent with the existing 'EXCLUDE MINORS from the training pool' guardrail ? or obtain Rule-10 verified parental consent and keep only de-identified object/coordinate data. NON-ATTRIBUTABILITY ARCHITECTURE (the 'secret file' question): Non-attributability is a property of the (data, holder, environment) triple, not of the timeseries. A per-video trajectory/feature stream is effectively a FINGERPRINT (de Montjoye: 4 points re-identify 95% of 1.5M people; k-anonymity collapses on rich quasi-identifiers), so a deterministic detector re-run on a candidate video can correlate-match it to its source ? BUT only if a correlatable reference EXISTS. PUBLIC source: a permanent cheap reference exists, re-linkage is 'reasonably likely', so a raw public-sourced stream should be treated as ATTRIBUTABLE; only heavy aggregation, perturbation/differential privacy, synthetic data, or a non-public enclave (NIST SP 800-188) breaks correlatability. PRIVATE/owner-recorded-and-unpublished source: there is no external reference to correlate against, so source-linkage by an EXTERNAL adversary is not 'reasonably likely' ? BUT it is WRONG to call this 'anonymous by default' (EDPS v. SRB, CJEU C-413/23 P, 4 Sept 2025): to the holder who can re-run the detector, it REMAINS personal data; it is merely RELATIVELY non-attributable to outsiders, and that protection is conditional, revocable (publish or leak the source and the link snaps back), and forward-looking (regulators say assume the data environment grows). HARD ENGINEERING IMPLICATION: the committed fixtures must NOT carry the source filename, match metadata, or the video_id (which is the MD5 of the source file ? using it as a public key would FINGERPRINT/confirm the source the instant anyone holds the file). Use an opaque, random, non-reversible surrogate id per record. Strip the 'name' field in *_golden_features.json. The lead's established constraint is correct: a contributor who RUNS the tests can necessarily read the plaintext fixtures and any key on their machine (DRM/analog-hole limit), so secrecy-from-the-contributor is impossible; the achievable goal is NON-ATTRIBUTABILITY + uselessness-in-isolation, with encryption useful ONLY as a barrier against the no-key public, never against a running contributor.",

"conditionsForSafePublicCommit": [

"PROVENANCE GATE: commit ONLY derived data from owner-recorded / fully-permissioned source video (Fork A). Tag every golden-set record with a verified provenance flag at ingestion; exclude any third-party-sourced (broadcast/YouTube/scraped/downloaded) record from the public commit. For owner-SHOT footage of a third party's event, first clear the admission ticket / venue / organiser / broadcaster terms ? only owner's own non-ticketed sessions are unconditionally clean.",

"SOURCE-LICENCE/ToS CHECK: for every source that touched the committed corpus, confirm in writing that its licence/ToS does NOT bar automated extraction OR redistribution of derived data. Do not rely on US fair use for the extraction step ? it is now contested (Thomson Reuters v. Ross; USCO May-2025) and unlawful acquisition independently defeats it (Bartz v. Anthropic). Redistribution (the public commit) is the single highest-risk act.",

"MINORS EXCLUSION: exclude all footage containing minors (under 18, per DPDP s.2(f)) from any persisted/committed derived stream beyond inanimate shuttle/interval data ? unless Rule-10 verifiable parental consent is held AND only de-identified object/coordinate data is kept. s.9(3)'s behavioural-monitoring ban means a persistent per-player pose/gait signature on a minor is prohibited.",

"BIOMETRIC EXCLUSION: do NOT publish pose keypoints, skeleton tracks, gait signatures, face crops, identifiable imagery, or any named-player-to-movement-signature mapping. Restrict the committed feature columns to shuttle (x,y)+Visibility, abstract rally/GT time-intervals, and non-biometric audio/court/aggregate-person features used only for rally detection and detached from identity.",

"DE-ATTRIBUTION: strip every source-identifying field from the committed fixtures ? source filename, match/event/player metadata, capture dates, and especially the video_id (MD5 of the source file). Replace with an opaque, random, non-reversible surrogate record id. The 'name' field in *_golden_features.json must be removed/aliased.",

"NON-ATTRIBUTABILITY POSTURE: because per-video trajectories are fingerprints correlatable to a source video, achieve non-attributability by keeping the SOURCE private (Fork A, unpublished) ? which makes external source-linkage not 'reasonably likely' ? and/or by coarsening/aggregating/perturbing where feasible. Document the (data, holder, environment) assessment and re-test it as the public-video environment grows (ICO/OVIC forward-looking duty). Treat the streams as personal data IN THE OWNER'S OWN HANDS regardless (EDPS v. SRB).",

"USELESS-IN-ISOLATION + ENCRYPTION-AS-PUBLIC-BARRIER ONLY: keep the committed fixtures useless without the regression harness + schema (abstract intervals + opaque ids, no imagery, no source pointer). Use encryption only as a barrier against the no-key public; do not represent it as protecting against a contributor who runs the tests (they can read plaintext + key locally ?

the lead's established DRM/analog-hole constraint).",

"PRECEDENT-CONSISTENT PLACEMENT: publish via the already-committed eval_baselines/ pattern (aggregate, provenance-agnostic JSON snapshots) for headline regression baselines; keep Fork-B-tainted or biometric-bearing *_golden_features.json private (per docs/GOLDEN_DATA_SHARING.md) and publish only a Fork-A, de-attributed, minors-excluded, biometric-free subset of features needed to run the suite.",

"DPDP HYGIENE: ensure any person-derived numeric column that could single out a player is genuinely identity-detached; do not rely on the DPDP s.3(c)(ii) 'publicly available data' carve-out for non-anonymized personal data (it is contested). Maintain purpose-limitation: features persisted for rally detection must not be repurposed for player identification.",

"COUNSEL SIGN-OFF: obtain retained India counsel (DPDP s.9/Rules, IT Act s.43/s.66, breach-of-confidence) AND EU/UK counsel (sui generis database right 'obtained vs created', CV-Online harm test, Ryanair contract) review of the final committed subset and the source-provenance log BEFORE the first public commit. This memo is general information, not advice."

],

"redFlags": [

"OVERCONFIDENCE TRAP #1 ? 'we created the data, so the EU/UK database right cannot apply.' Football Dataco v. Sportradar held live match-event data was OBTAINED (pre-existing on-court facts the recorder did not control), exactly analogous to a recorded badminton rally/shuttle position. A rights-holder can credibly argue the sui generis right subsists in our self-derived streams. Do not bank on the 'created' characterisation.",

"OVERCONFIDENCE TRAP #2 ? 'analytic extraction is strongly favorable fair use (US).' Thomson Reuters v. Ross (2025) REJECTED the 'copy an expressive work to reach unprotected facts' theory and found Factor-4 harm in a derivative data market; the USCO (May 2025) rejected 'non-expressive = inherently transformative.' If the dataset competes with a rights-holder's data/analytics market, fair use may fail. The extraction step is litigable, not settled.",

"OVERCONFIDENCE TRAP #3 ? 'owner's own recordings have no contract problem.' False for footage shot at a third party's event: the admission ticket is a contract barring recording/redistribution, and venue/organiser/broadcaster rights (BWF/league/Rome Convention) attach independently. Only the owner's own non-ticketed sessions are unconditionally clean.",

"OVERCONFIDENCE TRAP #4 ? 'private/unpublished source = anonymous by default.' Per EDPS v. SRB (CJEU 2025) the stream REMAINS personal data to the holder who can re-run the detector; it is only RELATIVELY non-attributable to an external adversary lacking the source, and that is conditional, revocable, and forward-looking. Misclassifying it as 'anonymous' is the exact GDPR error the controller's-perspective holding exists to prevent.",

"FINGERPRINTING / video_id leak: the project's video_id is the MD5 of the source file. Committing it (or any source filename/match metadata) as a public key would let anyone holding the candidate video CONFIRM the source instantly ? defeating non-attributability. Must be stripped and replaced with an opaque surrogate.",

"REDISTRIBUTION REMEDY is not nominal damages: the operative relief in scraping/ToS disputes is INJUNCTION + FORCED DELETION of the derived dataset (hiQ ordered destruction of all scraped-derived code/data). A public commit that breaches a source ToS can be ordered taken down and deleted ? a real, not theoretical, downside.",

"POSE/GAIT mislabeled low-risk: skeleton 'anonymization' leaves gait identity intact (80% re-ID), so any pose/gait stream is NOT reliably anonymous and is Art.9-adjacent. Excluding it is not optional polish ? it is a guardrail.",

"MINORS + behavioural monitoring: DPDP s.9(3) bars behavioural monitoring of under-18s and the Rule 12/Fourth-Schedule exemptions do not cover this project. A persistent per-player pose/gait signature on a minor is, conservatively, a prohibited act with criminal-adjacent regulatory exposure ? exclude minors from any persisted stream beyond inanimate shuttle/interval data.",

"MIXED-PROVENANCE corpus: the golden set may blend owned and third-party footage. A single Fork-B-tainted record contaminates the public commit. Provenance must be verified per-record, not assumed corpus-wide.",

"WRONG-LAW trade-secret framing: analysing the owner's moat under US 18 USC 1839 (DTSA) is a jurisdictional error ? the Bangalore owner's protection runs through Indian breach-of-confidence / Contract Act 1872 / IT Act s.72. The publish-forfeits-the-moat intuition survives, but the operative doctrine is different."

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"counselQuestions": [

"EU/UK sui generis database right: on the Football Dataco v. Sportradar 'obtained vs created' reasoning, does the right subsist in our self-derived shuttle-coordinate / rally-event streams, and if so does redistributing a de-attributed regression subset 'extract/re-utilise a substantial part' and cause CV-Online-Latvia-type harm to any rights-holder's feed? (EU AND UK, given likely English-language market.)",

"US fair use for the EXTRACTION step, post-Thomson Reuters v. Ross and the USCO May-2025 report: is transient analytic copying of a source video to compute rally/coordinate features defensible where the resulting dataset could be seen to compete in a 'data/analytics' market ? and does it matter that we only ever publish numbers, never the footage?",

"India: are anti-scraping/ToS terms enforceable in Indian courts, and what is the realistic IT Act s.43 (civil) / s.66 (criminal) exposure for ingesting third-party footage to derive features ? and does the pending ANI v. OpenAI judgment (reserved, ~April 2026) change the fair-dealing/TDM analysis?",

"India DPDP s.9 + Rules 10/12: for a corpus that may contain players under 18, what verifiable-parental-consent and exemption analysis applies, and does a persistent per-player pose/gait (or even per-player rally-tracking) feature constitute prohibited 'behavioural monitoring/tracking' of a child under s.9(3)?",

"Owner-shot event footage: do the admission tickets / venue / organiser (e.g., BWF, local league) / broadcaster terms for the specific tournaments and clubs in our corpus bar recording, derivation, or redistribution of derived data ? and which records must therefore be excluded from the public commit?",

"Personal-data classification under GDPR + DPDP for the SPECIFIC committed feature columns: at what point do identity-detached person/court/coordinate features 'single out' a player (especially in a small known squad/roster), and does our de-attribution + opaque-surrogate-id scheme suffice to keep the published subset outside personal-data obligations for downstream recipients (per EDPS v. SRB)?",

"Trade-secret / breach-of-confidence (Indian law): given commercialization and possible paper co-authorship, what is the moat trade-off of publishing the Fork-A subset, and are there confidentiality/Contract-Act 1872 / IT Act s.72 steps to take before forfeiting secrecy on any owner-recorded derived data?",

"Source-licence audit: for each source category in the corpus (owner-recorded, permissioned partner, downloaded/scraped), confirm the licence/ToS expressly permits BOTH automated extraction AND public redistribution of derived numeric data ? and where silent or prohibitive, which records are ineligible.",

"Encryption + 'secret file' design: does adding encryption that a running contributor can necessarily decrypt create any misrepresentation/false-security or DPDP 'reasonable security safeguards' compliance question, and is non-attributability-via-private-source a defensible posture to assert to regulators given its conditional/revocable nature?",

"Cross-border data transfer: does committing any record that is (or may be argued to be) personal data to a global public GitHub repo implicate DPDP cross-border-transfer rules or GDPR international-transfer obligations, independent of the copyright/contract analysis?"

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  },
  {
    "name": "Sweeney, k-Anonymity: A Model for Protecting Privacy (EPIC mirror)",
    "url": "https://epic.org/wp-content/uploads/privacy/reidentification/Sweeney_Article.pdf"
  },
  {
    "name": "de Montjoye et al., Unique in the Crowd (Scientific Reports 2013) ? 4 points ->
95%",
    "url": "https://dspace.mit.edu/bitstream/handle/1721.1/88233/de%20Montjoye-2013-
Unique%20in%20the%20crowd.pdf?sequence=1&isAllowed=y"
  },
  {
    "name": "OVIC ? The Limitations of De-Identification (data environment / external linkage;
forward-looking)",
    "url": "https://ovic.vic.gov.au/privacy/resources-for-organisations/the-limitations-of-de-
identification-protecting-unit-record-level-personal-information/"
  },
  {
    "name": "Trajectory-Based Spatiotemporal Entity Linking (arXiv 2010.01516)",
    "url": "https://arxiv.org/pdf/2010.01516"
  }
]
}

```

2. assistant (2026-06-16T03:54:46.732Z)

I'll attack this combined recommendation as a hostile red-teamer. Let me first ground myself in the actual code referenced, since the design makes many load-bearing claims about specific files and line numbers that, if wrong, collapse parts of the architecture.

3. user (2026-06-16T03:54:50.273Z)

Already up to date.

---BRANCH---

master

960c00d docs: rally-quality research + eval-framework roadmap (foundation doc) (#184)

4. user (2026-06-16T03:54:50.406Z)

```

backend\eval\__init__.py
backend\eval\classifier.py
backend\eval\training.py
backend\eval\gemini_refine.py
backend\eval\export_wasb_segments.py
backend\eval\metrics.py
backend\eval\regression.py
backend\eval\annotations.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py

```

```
backend\eval\rally_seq_proto.py
backend\eval\calibrate_local.py
backend\eval\gt_loader.py
backend\eval\harness.py
backend\eval\eval_stats.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
backend\eval\fusion_audio.py
backend\eval\fusion_compare.py
backend\eval\experiment.py
backend\eval\batch.py
backend\eval\fusion_golden.py
backend\eval\per_user_gate.py
backend\eval\per_user_eval.py
backend\eval\per_user_regression.py
backend\eval\ablation.py
backend\eval\served_gate_a.py
backend\eval\distill_local.py
backend\eval\promotion.py
backend\eval\serve_contrast.py
backend\eval\windowing.py
backend\eval\splits.py
```

5. assistant (2026-06-16T03:54:53.155Z)

Let me read the core files the design's claims depend on. I'll verify the line-number references and data contracts in parallel.

6. user (2026-06-16T03:54:54.110Z)

```
1      """Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden
set.
2
3      The fusion-P3 measurement. Loads the per-video feature JSONs written by
`fusion_golden.py`
4      and runs the SHIPPED experiment harness (`experiment.compare`, honest leave-one-video-
out) ?
5      no new scoring/labeling math. The two variants differ ONLY by `feature_names` (the
harness's
6      designed seam): the person features are extra columns the learned LogisticRegression
weights,
7      so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.
8
9      Run (no GPU):
10         python -m backend.eval.fusion_compare --features-dir output --record
11         """
12         from __future__ import annotations
13
14         import argparse
15         import glob
16         import json
17         import os
18         from typing import Any, Dict, List
19
20         from backend.eval import experiment
21         from backend.eval.eval_stats import paired_delta_ci
22         from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES
23
24
25         def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
26             """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign
test
27             (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out
scores
28             by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-
```

```

videos."""
29     by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
30     by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
31     vids = [p["video"] for p in res["variant_a"]["per_video"]]
32     return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])
33
34
35 def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
36     """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
37     feature columns + golden gts). The harness derives labels from gts itself."""
38     videos: List[experiment.VideoCandidates] = []
39     for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
40         with open(path, encoding="utf-8") as f:
41             d = json.load(f)
42             cands = d["candidates"]
43             intervals = [(float(c["start"]), float(c["end"])) for c in cands]
44             features: Dict[str, List[float]] = {}
45             for fn in SHUTTLE_FEATURE_NAMES:
46                 features[fn] = [float(c["shuttle"][fn]) for c in cands]
47             for fn in PERSON_FEATURE_NAMES:
48                 features[fn] = [float(c["person"][fn]) for c in cands]
49             gts = [(float(a), float(b)) for a, b in d["gts"]]
50             videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
51                                                         features=features, gts=gts))
52     return videos
53
54
55 def compare(videos: List[experiment.VideoCandidates], iou: float = 0.5,
56             threshold: float = 0.5) -> Dict[str, Any]:
57     shuttle_only = experiment.Variant(name="shuttle_only",
58                                       feature_names=list(SHUTTLE_FEATURE_NAMES),
59                                       threshold=threshold)
60     fusion = experiment.Variant(name="shuttle_person_fusion",
61                                 feature_names=list(SHUTTLE_FEATURE_NAMES) +
62                                 list(PERSON_FEATURE_NAMES),
63                                 threshold=threshold)
64     return experiment.compare(videos, shuttle_only, fusion, iou=iou, honest=True)
65
66
67 def format_result(res: Dict[str, Any], iou: float) -> str:
68     a, b, d = res["variant_a"]["aggregate"], res["variant_b"]["aggregate"], res["delta"]
69     ds = delta_stats(res)
70     st = ds["sign_test"]
71     lines = [
72         f"=== Fusion P3 LOVO (n={res['num_videos']} golden videos, IoU>={iou}, honest)
73         ===",
74         f"    shuttle-only : F1={a['f1']:.3f}  P={a['precision']:.3f}  R={a['recall']:.3f}
75         IoU={a['mean_iou']:.3f}",
76         f"    +person      : F1={b['f1']:.3f}  P={b['precision']:.3f}  R={b['recall']:.3f}
77         IoU={b['mean_iou']:.3f}",
78         f"    ?F1 = {d['f1']:+.3f}  ({res['delta_pct']['f1']:+.1f}%)
79         ?P={d['precision']:+.3f}  ?R={d['recall']:+.3f}",
80         f"    paired ?F1 {ds['mean_delta']:+.3f}  [95% CI {ds['ci_low']:+.3f},
81         {ds['ci_high']:+.3f}] "
82         f"| sign test {int(st['wins'])}W/{int(st['losses'])}L (p={st['p_value']:.3f}) "
83         f"| CI clears 0: {ds['ci_excludes_zero']}",
84         "    per-video ?F1:",
85     ]
86     for pv in res["per_video_delta"]:
87         lines.append(f"        {pv['video']}<26s ?F1={pv['f1']:+.3f}")
88     return "\n".join(lines)
89
90
91 def main() -> None:
92     ap = argparse.ArgumentParser(description="LOVO fusion lift: shuttle-only vs

```

```

shuttle+person (golden set).")
86     ap.add_argument("--features-dir", default="output")
87     ap.add_argument("--iou", type=float, default=0.5)
88     ap.add_argument("--record", action="store_true", help="append the result to the
experiment leaderboard")
89     args = ap.parse_args()
90
91     videos = load_videos(args.features_dir)
92     if len(videos) < 2:
93         raise SystemExit(f"need >=2 *_golden_features.json in {args.features_dir} (found
{len(videos)})")
94     res = compare(videos, iou=args.iou)
95     print(format_result(res, args.iou))
96     if args.record:
97         row = experiment.record_experiment(res)
98         print(f"recorded -> eval_baselines/experiment_leaderboard.json
({row['timestamp']}")
99
100
101     if __name__ == "__main__":
102         main()
103

```

7. user (2026-06-16T03:54:54.587Z)

```

1     """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3     Mirrors `distill_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4     candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5     EXISTING golden trajectory CSVs (so **no WASB re-run**), needs **no Gemini and no DB**,
and
6     writes one small JSON per video. The LOVO comparison (`fusion_compare.py`) then runs
100% on
7     CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8     measured by the shipped experiment harness; person features enter only as feature
columns the
9     learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11     Run (on the GPU box):
12     python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14     Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15     `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16     """
17     from __future__ import annotations
18
19     import argparse
20     import json
21     import os
22     import time
23     from typing import Any, Dict, List, Optional
24
25     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
26     from backend.pipeline.detectors import rally_gate
27     from backend.pipeline.detectors import fusion_features as ff
28     from backend.eval import distill_local
29     from backend.eval.calibrate_wasb import load_golden_gts
30     from backend.eval.training import label_candidates
31     from backend.utils.run_telemetry import RunTelemetry
32

```

```

33     # The 5 fps/resolution-invariant shuttle features distill_local already computes.
34     SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
35     # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
36     PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
37
38
39     def _derive_video_path(spec: Dict[str, Any]) -> str:
40         """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
41         golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
42         convention)."""
43         if spec.get("video_path"):
44             return str(spec["video_path"])
45         labels = str(spec["labels"])
46         if labels.endswith(".rallies.csv"):
47             return labels[: -len(".rallies.csv")] + ".MP4"
48         raise ValueError(f"{spec.get('name')}: no video_path and labels {labels!r} isn't
49         *.rallies.csv")
50
51     def extract_video(spec: Dict[str, Any], detector: Optional[Any] = None,
52                      frame_skip: int = 3, iou: float = 0.5,
53                      log_every_sec: float = 15.0,
54                      telemetry: Optional[Any] = None) -> Dict[str, Any]:
55         """One golden video -> a feature record (shuttle + person features per candidate
56         window).
57         Returns `{name, fps, frame_width,
58         candidates:[{start,end,shuttle{},person{},label}], gts}`.
59         `detector` (anything with `detections_over_windows`) is injectable for GPU-free
60         tests;
61         when None a real `PersonDetector` is built (loads torchvision on first use).
62         `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health
63         snapshots
64         from the detection progress callback so a hard kill leaves a forensic runlog.
65         """
66         fps, fw = float(spec["fps"]), float(spec["frame_width"])
67         traj = str(spec["trajectory"])
68         points = TrackNetRunner.parse_trajectory_csv(traj)
69         intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
70         gts = load_golden_gts(str(spec["labels"]))
71         labels = label_candidates(intervals, gts, iou)
72         # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
73         estimate
74         # (parity with the shuttle net-crossing gate) rather than hard-coding an
75         orientation.
76         axis, net_px, _span = rally_gate.estimate_play_axis(points)
77         video_path = _derive_video_path(spec)
78
79         if detector is None:
80             from backend.pipeline.detectors.person_detector import PersonDetector
81             detector = PersonDetector({"indexing": {"use_gpu": True}})
82
83         # Single-pass person detection over ALL candidate windows (open the video ONCE, read
84         # forward ? no per-window seek). Then compute features per window (pure, no GPU).
85         n = len(intervals)
86         win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]
87         print(f"[fusion_golden] {spec['name']}: {n} windows ? person-detecting in one
88         pass...", flush=True)
89         t0 = time.monotonic()
90         state = {"last": t0}
91
92         def _progress(done: int, total: int) -> None:
93             if telemetry is not None:
94                 telemetry.snapshot(done, total, phase=spec["name"])
95             now = time.monotonic()

```

```

89         if now - state["last"] >= log_every_sec or done >= total:
90             el = now - t0
91             rate = done / el if el > 0 else 0.0
92             eta = (total - done) / rate if rate > 0 else 0.0
93             pct = 100 * done // total if total else 0
94             print(f"[fusion_golden]    {spec['name']}: detect frame {done}/{total}
({pct}%) "
95                     f"| {el:.0f}s elapsed | ETA {eta:.0f}s", flush=True)
96             state["last"] = now
97
98     per_window = detector.detections_over_windows(video_path, win_frames, frame_skip,
99                                                    progress_cb=_progress)
100
101     candidates: List[Dict[str, Any]] = []
102     for i, ((s, e), xs) in enumerate(zip(intervals, x_shuttle)):
103         det = per_window[i]
104         det_fw = det.get("frame_width") or fw
105         det_fh = det.get("frame_height") or 0.0
106         # Normalise the (pixel) net coord to 0-1 on the play axis, matching the foot-
point
107         # normalisation fusion_features uses (x/width, y/height).
108         if axis == "x":
109             net_norm = net_px / det_fw if det_fw > 0 else 0.5
110         else:
111             net_norm = net_px / det_fh if det_fh > 0 else 0.5
112         sf, ef = win_frames[i]
113         shuttle_in = [p for p in points if sf <= p.frame <= ef]
114         person = ff.compute_person_features(det["frames"], shuttle_in, det_fw, det_fh,
115                                             court_polygon=None, net=(axis, net_norm))
116         candidates.append({
117             "start": float(s), "end": float(e),
118             "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
119             "person": {k: float(v) for k, v in person.items()},
120             "label": int(labels[i]),
121         })
122     return {
123         "name": spec["name"], "fps": fps, "frame_width": fw,
124         "candidates": candidates,
125         "gts": [[float(a), float(b)] for a, b in gts],
126     }
127
128
129     def run(manifest_path: str, out_dir: str, frame_skip: int = 3, iou: float = 0.5,
130            fresh: bool = False, smallest_first: bool = False) -> List[str]:
131         with open(manifest_path, encoding="utf-8") as f:
132             specs = json.load(f)
133         if smallest_first:
134             # Process small videos first for quick feedback (the slow full match lands
last);
135             # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
row/frame).
136             specs = sorted(specs, key=lambda s: (os.path.getsize(s["trajectory"])
137                                                  if os.path.isfile(s.get("trajectory", ""))
else 0))
138         os.makedirs(out_dir, exist_ok=True)
139         # Run-telemetry black box: a long GPU run can be OS-killed with no traceback, so
keep a tiny
140         # rotating runlog of velocity + machine health for the post-mortem. Constructing it
must never
141         # break extraction, so guard the build and fall back to no telemetry on any error.
142         telem: Optional[RunTelemetry]
143         try:
144             telem = RunTelemetry(os.path.join(out_dir, "run_telemetry.jsonl"),
label="fusion_golden")
145         except Exception: # noqa: BLE001 - telemetry wiring must never break the run it

```

```

observes
146         telem = None
147     written: List[str] = []
148     for vi, spec in enumerate(specs, 1):
149         out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
150         # Resumable: skip a video whose JSON already exists (a re-run picks up where a
prior
151         # run left off instead of re-detecting). --fresh forces a redo.
152         if not fresh and os.path.isfile(out):
153             print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already
extracted "
154                 f"-> {out} (use --fresh to redo)", flush=True)
155             written.append(out)
156             continue
157         print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
flush=True)
158         t0 = time.monotonic()
159         if telem is not None:
160             telem.event("video_start", spec["name"])
161         try:
162             data = extract_video(spec, frame_skip=frame_skip, iou=iou, telemetry=telem)
163         except Exception as e: # noqa: BLE001 - record machine state at death, then re-
raise
164             if telem is not None:
165                 telem.event("error", spec["name"], err=str(e))
166             raise
167         n_pos = sum(c["label"] for c in data["candidates"])
168         with open(out, "w", encoding="utf-8") as f:
169             json.dump(data, f, indent=2)
170         if telem is not None:
171             telem.event("video_done", spec["name"],
172                        windows=len(data["candidates"]), positives=n_pos)
173         print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: DONE in
{time.monotonic() - t0:.0f}s "
174             f"| {len(data['candidates'])} windows ({n_pos} positive) |
{len(data['gts'])} golden rallies "
175             f"-> {out}", flush=True)
176         written.append(out)
177     return written
178
179
180     def main() -> None:
181         ap = argparse.ArgumentParser(description="Extract golden-set shuttle+person features
(fusion P3 GPU step).")
182         ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
183         ap.add_argument("--out-dir", default="output")
184         ap.add_argument("--frame-skip", type=int, default=3)
185         ap.add_argument("--iou", type=float, default=0.5)
186         ap.add_argument("--fresh", action="store_true", help="re-extract even if a JSON
already exists")
187         ap.add_argument("--smallest-first", action="store_true",
188                        help="process smaller videos first for quick feedback (LOVO is
order-independent)")
189         args = ap.parse_args()
190         run(args.manifest, args.out_dir, args.frame_skip, args.iou, fresh=args.fresh,
191            smallest_first=args.smallest_first)
192
193
194     if __name__ == "__main__":
195         main()
196

```

8. user (2026-06-16T03:54:55.010Z)

```

1     """Pure temporal interval-matching metrics for rally evaluation.

```

```

2
3 No I/O, no DB, no global state ? every function here is deterministic and unit
4 tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5 start < end. We score a list of *predicted* rally intervals against a list of
6 *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7 """
8
9 from dataclasses import dataclass, field
10 from typing import List, Tuple, Optional
11
12 Interval = Tuple[float, float]
13
14
15 def interval_iou(a: Interval, b: Interval) -> float:
16     """Temporal intersection-over-union of two intervals.
17
18     Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19     overlap / union, both measured as durations on the timeline.
20     """
21     a_start, a_end = a
22     b_start, b_end = b
23     if a_end <= a_start or b_end <= b_start:
24         return 0.0
25     inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26     if inter <= 0.0:
27         return 0.0
28     union = (a_end - a_start) + (b_end - b_start) - inter
29     return inter / union if union > 0 else 0.0
30
31
32 @dataclass
33 class Match:
34     """One matched predicted?ground-truth pair."""
35     pred_index: int
36     gt_index: int
37     iou: float
38
39
40 @dataclass
41 class MatchResult:
42     """Outcome of matching predictions to ground truth at a given IoU threshold."""
43     matches: List[Match] = field(default_factory=list)
44     unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45     unmatched_gt_indices: List[int] = field(default_factory=list)   # false negatives
46 (misses)
47
48 def match_intervals(preds: List[Interval], gts: List[Interval],
49                    iou_threshold: float = 0.5) -> MatchResult:
50     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52     Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53     are claimed highest-IoU first, and each prediction/GT can be used once. This
54     keeps a single prediction from being credited for two ground-truth rallies
55     (and vice versa).
56     """
57     candidates = []
58     for pi, p in enumerate(preds):
59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             # Require real overlap: at iou_threshold=0.0 a plain `iou >= threshold`
62             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
63             if iou > 0.0 and iou >= iou_threshold:
64                 candidates.append((iou, pi, gi))

```

```

65         # Highest IoU first; ties broken by indices for determinism.
66         candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
67
68         used_pred, used_gt = set(), set()
69         matches: List[Match] = []
70         for iou, pi, gi in candidates:
71             if pi in used_pred or gi in used_gt:
72                 continue
73             used_pred.add(pi)
74             used_gt.add(gi)
75             matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
76
77         unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]
78         unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
79         return MatchResult(matches=matches,
80                             unmatched_pred_indices=unmatched_pred,
81                             unmatched_gt_indices=unmatched_gt)
82
83
84     @dataclass
85     class Score:
86         """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
87         iou_threshold: float
88         num_pred: int
89         num_gt: int
90         true_positives: int
91         false_positives: int
92         false_negatives: int
93         precision: float
94         recall: float
95         f1: float
96         mean_iou: float          # over matched pairs (0.0 if none)
97         mean_start_error: float  # mean |pred_start - gt_start| over matches, seconds
98         mean_end_error: float    # mean |pred_end - gt_end| over matches, seconds
99
100         def as_dict(self) -> dict:
101             return self.__dict__.copy()
102
103
104     def score(preds: List[Interval], gts: List[Interval],
105              iou_threshold: float = 0.5,
106              match_result: Optional[MatchResult] = None) -> Score:
107         """Compute precision/recall/F1, mean matched IoU, and boundary errors.
108
109         Pass a precomputed `match_result` to avoid re-matching (e.g. when also
110         extracting the failure lists); otherwise matching is done here.
111         """
112         mr = match_result if match_result is not None else match_intervals(preds, gts,
113                                     iou_threshold)
114
115         tp = len(mr.matches)
116         fp = len(mr.unmatched_pred_indices)
117         fn = len(mr.unmatched_gt_indices)
118
119         precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
120         recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
121         f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
122         else 0.0
123
124         if mr.matches:
125             mean_iou = sum(m.iou for m in mr.matches) / tp
126             mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
127                                  mr.matches) / tp
128             mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
129                                 mr.matches) / tp

```

```

126         else:
127             mean_iou = mean_start_err = mean_end_err = 0.0
128
129         return Score(
130             iou_threshold=iou_threshold,
131             num_pred=len(preds), num_gt=len(gts),
132             true_positives=tp, false_positives=fp, false_negatives=fn,
133             precision=precision, recall=recall, f1=f1,
134             mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
135         )
136
137
138     def pr_curve(preds: List[Interval], gts: List[Interval],
139                 thresholds: Optional[List[float]] = None) -> List[Score]:
140         """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).
141
142         Lets you see strict-vs-lenient behaviour rather than a single operating point.
143         """
144         if thresholds is None:
145             thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
146         return [score(preds, gts, iou_threshold=t) for t in thresholds]
147

```

9. assistant (2026-06-16T03:54:59.902Z)

Now I've confirmed key claims. The metrics are pure differences (translation-invariant, as claimed), and `fusion\_compare.load\_videos` reads `c["shuttle"][fn]` and `c["person"][fn]`. Critically, I note `load\_videos` ***unconditionally reads the person block*** ? that's a real issue for the Tier-0 path. Let me verify the remaining load-bearing claims: the committed leak file, the skipif pattern, regression.py, and run_all_tests.

10. user (2026-06-16T03:55:01.984Z)

```

total 28
drwxr-xr-x 1 avidu 197609    0 Jun 14 16:28 .
drwxr-xr-x 1 avidu 197609    0 Jun 16 01:52 ..
-rw-r--r-- 1 avidu 197609 4696 Jun 14 16:28 experiment_leaderboard.json
-rw-r--r-- 1 avidu 197609 1977 Jun 10 17:43 gemini_wrapper_2026-06-10.json
-rw-r--r-- 1 avidu 197609  689 Jun 10 15:57 heuristic_lovo_n6.json
===REGRESSION HEAD===
"""Regression runner ? verify the production tool against golden-set ground truth.

```

Implements the explicit ask: **"given a video, get a golden set created and verify its precision and recall against the production tool."** This is deliberately a thin wrapper over machinery that already exists:

ground truth	? AnnotationProvider (file / vendor / oracle)	[backend.annotations]
predictions	? the production pipeline's stored DB rows	[backend.eval.harness]
comparison	? interval matching + P/R/F1	[backend.eval.metrics]
baseline	? a per-video snapshot of "known good" numbers	[this module]

A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI), later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.

```

"""
import os
import json
import hashlib
import logging
from dataclasses import dataclass, asdict
from typing import Dict, List, Optional

from backend.infrastructure.database import Database
from backend.eval import metrics

```

```
from backend.eval.harness import get_predictions
from backend.annotations import annotation_registry
```

```
logger = logging.getLogger(__name__)
```

Tracked location (NOT under the gitignored output/), so a fresh checkout / CI run can actually load a committed baseline to compare against.

```
DEFAULT_BASELINE_PATH = os.path.join("eval_baselines", "regression_baseline.json")
DEFAULT_TOLERANCE = 0.05
```

```
def gt_hash(ground_truth: List[metrics.Interval]) -> str:
    """Stable short hash of the GT intervals ? detects a baseline taken against different
    labels."""
    norm = sorted((round(float(s), 3), round(float(e), 3)) for s, e in ground_truth)
    return hashlib.sha1(json.dumps(norm).encode()).hexdigest()[:12]
```

```
@dataclass
class RegressionResult:
    video_id: str
```

11. user (2026-06-16T03:55:02.738Z)

```
1      {
2      "name": "heuristic_velocity_windowing_LOVO",
3      "description": "The honest cross-video floor the Gen-0 learned model must beat:
calibrated velocity-windowing heuristic, leave-one-video-out over the n=6 owned golden corpus.",
4      "computed": "2026-06-10, reproduced via: python -m backend.eval.calibrate_local
--manifest output/human_lovo_manifest.json --metric f1 (matches docs/archives/quality-
iterations/2026-06-multivideo-lovo)",
5      "mean_f1": 0.480,
6      "std": 0.242,
7      "per_video": {
8      "adarsh_avi_singles": 0.751,
9      "kushagra_singles": 0.157,
10     "largetest_doubles": 0.737,
11     "gbaaddy": 0.309,
12     "testlarge_short": 0.267,
13     "mahadevpura_singles": 0.657
14     }
15     }
16
```

12. user (2026-06-16T03:55:03.519Z)

```
360         Signal("shuttle.duration", "shuttle", ("duration",)),
361         Signal("shuttle.velocity", "shuttle", ("peak_velocity", "mean_velocity",
"active_ratio")),
362         Signal("gate_a", "gate_a", ("net_crossings",), gate_knob="min_crossings",
structural=True),
363         Signal("person", "person", PERSON_COLUMNS),
364         Signal("audio", "audio", AUDIO_COLUMNS),
365     ]
366
367
368     def static_gate_a_signal() -> Signal:
369         """The Gate-A-only structural signal (a pure gate knob over the recall-oriented
proposal
370         set). Ablating it against the static ``Variant(min_crossings=2)`` full reproduces
the
371         hand-measured Gate-A result (?F1 ? +0.074, 6/6, p ? 0.031) ? the framework's litmus
test."""
```

```

372         return Signal("gate_a", "gate_a", (), gate_knob="min_crossings", structural=True)
373
374
375     def litmus_config() -> AblationConfig:
376         """The Gate-A reproduction config: full = STATIC ``Variant(min_crossings=2)`` (no
learned
377         classifier), inventory = the single Gate-A gate signal. Ablating it = CONTROL+gate
vs CONTROL
378         ? exactly the historical measurement (``scratch/gate_ab_ablation.py``)."""
379         full = experiment.Variant(name="control_gate_a", min_crossings=2)
380         return AblationConfig(full=full, signals=[static_gate_a_signal()],
granularity="signal")
381
382
383     # ----- corpus
I/O
384
385
386     def load_videos(features_dir: str) -> List[VideoCandidates]:
387         """Load every ``*_golden_features.json`` into a ``VideoCandidates`` (intervals +
shuttle +
388         person + audio columns + golden gts). Audio columns are optional (older JSONs
predate the
389         audio augment) ? a missing block defaults those columns to 0.0 so the loader never
crashes,
390         mirroring ``experiment._feature_column``'s tolerance."""
391         videos: List[VideoCandidates] = []
392         for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
393             with open(path, encoding="utf-8") as f:
394                 d = json.load(f)
395                 cands = d["candidates"]
396                 intervals = [(float(c["start"]), float(c["end"])) for c in cands]
397                 features: Dict[str, List[float]] = {}
398                 for fn in SHUTTLE_COLUMNS:
399                     features[fn] = [float(c["shuttle"][fn]) for c in cands]
400                 for fn in PERSON_COLUMNS:
401                     features[fn] = [float(c["person"][fn]) for c in cands]
402                 for fn in AUDIO_COLUMNS:
403                     features[fn] = [float((c.get("audio") or {}).get(fn, 0.0)) for c in cands]
404                 gts = [(float(a), float(b)) for a, b in d["gts"]]
405                 videos.append(VideoCandidates(name=d["name"], intervals=intervals,
406                                             features=features, gts=gts))
407         return videos
408
409
410     def learned_full_variant(*, threshold: float = 0.5, min_crossings: int = 0) -> Variant:
411         """The learned full-set reference: all 13 columns (shuttle 5 + person 5 + audio 3)
weighted
412         by the LOVO logistic scorer. Used for the LOSO/LOGO contribution table over the
learned path
413         (distinct from the static Gate-A litmus)."""
414         cols = list(SHUTTLE_COLUMNS) + list(PERSON_COLUMNS) + list(AUDIO_COLUMNS)
415         return experiment.Variant(name="full_learned", feature_names=cols,
416                                 threshold=threshold, min_crossings=min_crossings)
417
418
419     # ----- table
rendering

```

13. assistant (2026-06-16T03:55:08.053Z)

This is critical. The committed `heuristic\_lovo\_n6.json` file confirms the live leak: `adarsh\_avi\_singles`, `kushagra\_singles`, `gbaaddy`, `mahadevpura\_singles` are real player/source names committed to git. The design's "COMMITTED-LEAK REMEDIATION" correctly identifies this. But note this file is ****already in git**** ? let me confirm it's tracked, check

the skipif pattern, and verify the `\_feature\_column` 0.0-default claim (load-bearing for the additive-column scheme).

14. user (2026-06-16T03:55:11.376Z)

```
eval_baselines/experiment_leaderboard.json
eval_baselines/gemini_wrapper_2026-06-10.json
eval_baselines/heuristic_lovo_n6.json
===LOG OF LEAK FILE===
d51efe3 feat(training): Gen-0 TAL harness ? one-command train -> honest LOVO -> ship-gate ->
artifact (M0.4 scaffold) (#77)
===IS mahadevpura ANYWHERE ELSE TRACKED?===
backend/eval/eval_partial_labels.py
eval_baselines/gemini_wrapper_2026-06-10.json
eval_baselines/heuristic_lovo_n6.json
tests/test_cleanup_caches.py
```

15. user (2026-06-16T03:55:11.913Z)

```
360      # ----- the
golden litmus
361
362      GOLDEN_DIR = os.path.join(
363          os.path.dirname(os.path.abspath(__file__)), "output")
364
365
366      def _golden_present() -> bool:
367          return len(glob.glob(os.path.join(GOLDEN_DIR, "*_golden_features.json"))) >= 6
368
369
370      @pytest.mark.skipif(not _golden_present(),
371                          reason="golden output/*_golden_features.json absent (run
fusion_golden first)")
372      def test_litmus_reproduces_gate_a():
373          """LITMUS: on the 6 real golden feature JSONs the runner reproduces the hand-
measured Gate-A
374          result ? marginal ?F1 ? +0.074, 6/6 wins, p ? 0.031, CI ? [+0.042, +0.104], within
tolerance.
375          This validates the whole framework against the recorded ground truth. Skips cleanly
in CI
376          where the golden substrate is absent; runs on the owner box where it is present."""
377          videos = ablation.load_videos(GOLDEN_DIR)
378          assert len(videos) == 6
379          cfg = ablation.litmus_config()
380          report = run_ablation(videos, cfg)
381          assert len(report.rows) == 1
382          row = report.rows[0]
383          assert row.signal == "gate_a"
384          # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within tolerance.
```

16. assistant (2026-06-16T03:55:16.992Z)

The leak is confirmed in git history (commit d51efe3, PR #77) and also appears in `eval\_partial\_labels.py`, `gemini\_wrapper\_2026-06-10.json`, and test files. This is important: the design's remediation only scrubs `heuristic\_lovo\_n6.json`, but the names leak in **multiple committed files** and are **already in immutable git history**. Let me check the additional leak sites and the `\_feature\_column` default claim.

17. user (2026-06-16T03:55:20.587Z)

```
===eval_partial_labels===
backend/eval/eval_partial_labels.py:23:      --trajectory
C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
backend/eval/eval_partial_labels.py:119:      comp_out = comparison_out or osp.join("output",
```

```

"adarsh_avi_vs_human.csv")
===gemini_wrapper===
eval_baselines/gemini_wrapper_2026-06-10.json:7:      "mahadevpura_singles": {
eval_baselines/gemini_wrapper_2026-06-10.json:12:     "kushagra_singles": {
eval_baselines/gemini_wrapper_2026-06-10.json:19:     "heuristic_lovo": {"mahadevpura_singles":
0.657, "kushagra_singles": 0.157},
eval_baselines/gemini_wrapper_2026-06-10.json:20:     "gen0_learned_lovo":
{"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}
===test_cleanup===
tests/test_cleanup_caches.py:19:     assert classify("adarsh_traj.csv", False, "wsl")[2] is False
# trajectory -> keep
tests/test_cleanup_caches.py:24:     cat, regen, default_purge =
classify("adarsh_traj_wasbcache", True, "wsl")
===any 32-hex MD5 already committed (video_id)?===

```

18. assistant (2026-06-16T03:55:24.541Z)

Confirmed: the player-name leak exists in at least **four committed files** plus git history, not just the one the design names. Now let me verify the ``_feature_column` 0.0`-default behavior and ``experiment.predict/compare``, plus the ``VideoCandidates`` schema, since the design's "additive column / Tier-0 strips person" rests on the harness tolerating absent columns.

19. user (2026-06-16T03:55:25.999Z)

```

140         "calibrate": self.calibrate,
141     }
142
143     @classmethod
144     def from_dict(cls, d: dict) -> "Variant":
145         known = {f for f in cls.__dataclass_fields__} # type: ignore[attr-defined]
146         return cls(**{k: v for k, v in d.items() if k in known})
147
148     def spec_hash(self) -> str:
149         """Stable 12-char hash of the recipe ? identifies the variant in the leaderboard
150         and makes a result reproducible. The pinned control variant always hashes
151         the
152         same, so a regression is always traceable to a recipe change, never to drift."""
153         blob = json.dumps(self.to_dict(), sort_keys=True, default=str)
154         return sha1(blob.encode()).hexdigest()[:12]
155
156     #: The pinned, frozen baseline: candidates as detected, merged, no gate, no learned
157     #: filter. Always the comparison reference; "rollback" = make this the served variant.
158     CONTROL = Variant(name="control")
159
160
161     # ----- persisted candidates
162
163
164     @dataclass
165     class VideoCandidates:
166         """One video's persisted detection, ready for offline re-scoring.
167
168         ``intervals`` are the candidate windows; ``features`` is column-major
169         (``feature_name -> per-candidate value``, each list aligned to ``intervals``);
170         ``gts`` are golden labels (None for new/unlabeled footage). Build one from the DB
171         with ``load_video_candidates``, or directly in tests.
172         """
173
174         name: str
175         intervals: List[Interval]
176         features: Dict[str, List[float]] = field(default_factory=dict)
177         gts: Optional[List[Interval]] = None
178
179

```

```

180     def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
181         """Persisted feature column, defaulting missing/short columns to 0.0 (matches
182         `training.extract_yolo_features`, which lets a sparse/old row still vectorise)."""
183         col = vc.features.get(name)
184         n = len(vc.intervals)
185         if col is None:
186             logger.warning("variant references feature %r absent from %s ? treating as 0.0",
187                             name, vc.name)
188             return [0.0] * n
189         if len(col) != n:
190             raise ExperimentError(
191                 f"feature {name!r} has {len(col)} values but {vc.name} has {n} candidates")
192         return [float(x) for x in col]
193
194
195     def _feature_matrix(vc: VideoCandidates, feature_names: Sequence[str]) ->
List[List[float]]:
196         cols = [_feature_column(vc, name) for name in feature_names]
197         return [list(row) for row in zip(*cols)] if cols else [[] for _ in vc.intervals]
198
199

```

20. user (2026-06-16T03:55:27.918Z)

===run_all_tests glob===

4:A single, discoverable entry point so you (or a reviewer / CI) can verify the

===how does run_all_tests find tests?===

#!/usr/bin/env python

"""Run the FULL test suite for the badminton-highlight-indexer.

A single, discoverable entry point so you (or a reviewer / CI) can verify the whole suite before opening a PR:

```

python run_all_tests.py           # full suite, quiet
python run_all_tests.py -v        # verbose (passthrough to pytest)
python run_all_tests.py -k config # filter by keyword (passthrough)
python run_all_tests.py --cov=backend --cov-report=term-missing # coverage

```

Anything after the script name is passed straight through to pytest.

Notes

- The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
- The observability **reporting** tests skip cleanly if the optional `obsreport` package isn't installed (they use `pytest.importorskip`). To exercise them too:

```
pip install -e ../sports-obsreport --no-build-isolation
```
- Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).

"""

from __future__ import annotations

import importlib.util

import subprocess

import sys

def _installed(module: str) -> bool:

```

    try:
        return importlib.util.find_spec(module) is not None
    except (ImportError, ValueError):
        return False

```

def main() -> int:

```

    args = list(sys.argv[1:])
    # Default to quiet unless the caller chose a verbosity.

```

```

if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
    args = ["-q", *args]
cmd = [sys.executable, "-m", "pytest", *args]

obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests will
SKIP"
bar = "=" * 72
print(bar)
print(" Full test suite: " + " ".join(cmd))
print(f" obsreport (optional reporting dep): {obs}")
print(bar, flush=True)

rc = subprocess.run(cmd).returncode

print(bar)
print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit {rc}"))
print(bar)
return rc

if __name__ == "__main__":
    raise SystemExit(main())

```

21. assistant (2026-06-16T03:55:33.794Z)

I have a critical finding here. The `\_feature\_column` for a missing column **logs a warning but returns 0.0 and proceeds** ? it does NOT raise. This is the load-bearing mechanism for the design's "additive column" tolerance, but it's a double-edged sword I'll attack below (silent wrong-answer mode). Let me now verify the determinism claims (float repr, `gt\_hash` rounding) and check the GOLDEN_DATA_SHARING doc and the existing test that builds a synthetic corpus, plus confirm there's no existing leak-scan/pre-commit infrastructure.

22. user (2026-06-16T03:55:37.915Z)

```

===GOLDEN_DATA_SHARING===
docs/GOLDEN_DATA_SHARING.md

```

Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions

****Status:**** convention + plan (2026-06-15). Tracked in ****#175****.
****Purpose:**** a single cloud-synced location for the GPU-extracted golden artifacts (trajectory CSVs, `\_golden\_features.json`, manifests, baseline snapshots) so the ****GPU box**** and ****CPU dev/agent sessions**** share the same files ? without committing large data to git.

Why this exists

The rally-quality measurement is split across two machines by capability:

- ****`backend/eval/fusion\_golden.py`**** (shuttle+person feature ****extraction****) needs the ****videos****, the ****trajectory CSVs****, the ****`.rallies.csv`** golden labels, and a ****GPU/WSL**** stack ? only the GPU box can run it.
- ****`backend/eval/fusion\_compare.py`**** / ****`backend/eval/ablation.py`**** (the LOVO ****litmus****: ?F1, bootstrap CI, paired sign test) are ****pure-CPU re-scoring**** that read only the small ****_golden_features.json`** ? any session (incl. a CPU dev container) can run them.

Today those artifacts sit in gitignored `output/` (per-machine) and scattered local folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap with OneDrive.

Canonical shared folder

```

`%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ? `C:\Users\avidu\OneDrive\rally-annotator\golden-data\`

```

OneDrive is present + signed-in on both the GPU box and the dev machine under the same account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an

```
overwrite):
```

```
...
```

```
golden-data/
```

```
  README.md
  features/<corpus-tag>/      *_golden_features.json    (small ? the key shared artifact)
  trajectories/<corpus-tag>/  *_predict.csv / *_wasb.csv (medium ? reusable across runs)
  manifests/                  *_lovo_manifest.json
  baselines/                  leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
...
```

`<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). ****Do NOT**** put raw videos here (tens of GB) ? keep those local to the GPU box.

The GPU ? CPU workflow

1. ****GPU box**** (videos + CSVs + labels present):

```
...
python -m backend.eval.fusion_golden --manifest <manifest> \
  --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"
...
```

(or extract to local `output/` then copy the `\*\_golden\_features.json` into the shared folder).

2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).

3. ****CPU dev/agent session**** points the analysis at the shared folder:

```
...
python -m backend.eval.fusion_compare --features-dir "...golden-data\features\2026-06-15-n6"
--record
python -m backend.eval.ablation      --features-dir "...golden-data\features\2026-06-15-n6"
--granularity group
===any pre-commit / hooks?===
ls: cannot access '.pre-commit-config.yaml': No such file or directory
ls: cannot access '.githubhooks/': No such file or directory
.github/workflows/:
total 4
drwxr-xr-x 1 avidu 197609    0 Jun 15 14:11 .
drwxr-xr-x 1 avidu 197609    0 Jun 10 11:31 ..
-rw-r--r-- 1 avidu 197609 4032 Jun 15 14:11 ci.yml
===_learned_corpus test builder===
tests/test_ablation.py:432:def _write_learned_corpus(d: str, n: int = 6) -> None:
tests/test_ablation.py:455:     _write_learned_corpus(d)
tests/test_ablation.py:473:     _write_learned_corpus(d)
tests/test_ablation.py:484:     _write_learned_corpus(d, n=1)
tests/test_experiment_honest.py:23:def _corpus():
tests/test_per_user_regression.py:47:def _corpus(n: int = 6):
Binary file tests/__pycache__/test_ablation.cpython-313-pytest-9.0.3.pyc matches
```

23. assistant (2026-06-16T03:55:42.814Z)

There is ****no existing pre-commit hook infrastructure**** ? the design's central guard (PRE-COMMIT + CI LEAK GUARD) must be built from scratch, and `.githubhooks/` is trivially bypassed (not enforced server-side). Let me read the CI workflow to see what the leak-scan would attach to, and verify the `\_write\_learned\_corpus` synthetic builder and the determinism-sensitive float handling. Then I'll examine `eval\_partial\_labels.py` line 23 (a hardcoded absolute path with a username ? itself a leak).

24. user (2026-06-16T03:55:43.770Z)

```
1      name: CI
2
3      on:
4        push:
5          branches: [master]
6        pull_request:
```

```

7
8 permissions:
9   contents: read
10
11 concurrency:
12   group: ci-${{ github.ref }}
13   cancel-in-progress: true
14
15 jobs:
16   # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17   # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
18   # tracked .py, including scripts/ and training/, which the backend-only
19   # py_compile smoke sweep does not reach.
20   lint:
21     name: Ruff lint
22     runs-on: ubuntu-latest
23     steps:
24       - uses: actions/checkout@v6
25       - uses: actions/setup-python@v6
26         with:
27           python-version: "3.13"
28       - name: Install ruff
29         run: python -m pip install ruff
30       - name: Ruff check
31         run: ruff check . --output-format github
32
33   # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
34   # per-module quarantine list ? ratchet by removing entries as modules clean
35   # up. Typed core deps are installed so mypy sees their real signatures
36   # (without them ignore_missing_imports would Any-stub half the checks).
37   typecheck:
38     name: Mypy (lenient baseline)
39     runs-on: ubuntu-latest
40     steps:
41       - uses: actions/checkout@v6
42       - uses: actions/setup-python@v6
43         with:
44           python-version: "3.13"
45       - name: Install mypy + typed core deps
46         run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-
47 genai numpy
48       - name: Mypy
49         run: mypy backend training
50
51   # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
52   # module-level import error shipping while the suite stays green.
53   # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
54   # on machines without the GPU stack (test_import_smoke stubs what's absent).
55   import-smoke:
56     name: Syntax sweep + import smoke (no GPU stack)
57     runs-on: ubuntu-latest
58     steps:
59       - uses: actions/checkout@v6
60       - uses: actions/setup-python@v6
61         with:
62           python-version: "3.13"
63       - name: Install minimal deps
64         run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
65 genai
66       - name: Run import smoke tests
67         run: python -m pytest tests/test_import_smoke.py -v
68
69   full-suite:
70     name: Full test suite (offline, mocked)
71     runs-on: ubuntu-latest

```

```

70     steps:
71     - uses: actions/checkout@v6
72     - uses: actions/setup-python@v6
73       with:
74         python-version: "3.13"
75         cache: pip
76     - name: System libs for opencv-python
77       run: |
78         sudo apt-get update
79         sudo apt-get install -y libgl1 libglu1-mesa-dev || sudo apt-get install -y
libgl1 libglu1-mesa-dev
80     - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81       # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82       # need an out-of-band index that PEP 621 can't express. Keep installing
83       # them separately here, before the editable install pulls the rest.
84       run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85     - name: Install package (editable) + dev tooling
86       # Editable PEP 621 install: runtime deps + the `dev` group
87       # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88       run: python -m pip install -e .[dev]
89       # obsreport (private, soft dep) is absent here: reporting tests skip via
90       # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91       # is therefore calibrated against THIS lane's number (local runs with
92       # obsreport installed measure a little higher).
93       # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94       # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95       # without an owner decision.
96     - name: Run full suite (with coverage floor)
97       run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
98

```

25. user (2026-06-16T03:55:44.428Z)

```

432     def _write_learned_corpus(d: str, n: int = 6) -> None:
433         for k in range(n):
434             cand = []
435             for (s, e), pos in [((0.0, 10.0), 1), ((20.0, 30.0), 1), ((40.0, 41.0), 0),
((50.0, 51.0), 0)]:
436                 shuttle = {fn: 1.0 for fn in ablation.SHUTTLE_COLUMNS}
437                 person = {fn: 0.0 for fn in ablation.PERSON_COLUMNS}
438                 person["players_in_court"] = 4.0 if pos else 0.0
439                 person["in_court_ratio"] = 1.0 if pos else 0.0
440                 audio = {fn: 0.0 for fn in ablation.AUDIO_COLUMNS}
441                 cand.append({"start": s, "end": e, "shuttle": shuttle, "person": person,
442                             "audio": audio, "label": pos})
443                 data = {"name": f"v{k}", "fps": 30.0, "frame_width": 1920.0,
444                         "candidates": cand, "gts": [[0.0, 10.0], [20.0, 30.0]]}
445                 with open(os.path.join(d, f"v{k}_golden_features.json"), "w", encoding="utf-8")
as f:
446                     json.dump(data, f)
447
448
449     def test_cli_main_writes_pdf_and_json(tmp_path, monkeypatch, capsys):
450         """`main()` runs the LOGO ablation and writes both a PDF and the structured
JSON."""
451         pytest.importorskip("reportlab")
452         pytest.importorskip("pypdf")
453         d = str(tmp_path / "feat")
454         os.makedirs(d)
455         _write_learned_corpus(d)
456         pdf = str(tmp_path / "out.pdf")
457         js = str(tmp_path / "out.json")
458         monkeypatch.setattr("sys.argv", ["ablation", "--features-dir", d, "--granularity",

```

```

"group",
459                                     "--pdf", pdf, "--json", js])
460     ablation.main()
461     out = capsys.readouterr().out
462     assert "Ablation report" in out
463     assert os.path.isfile(pdf) and os.path.getsize(pdf) > 1000
464     assert os.path.isfile(js)
465     report = json.loads(open(js, encoding="utf-8").read())
466     assert {r["group"] for r in report["rows"]} == {"shuttle", "gate_a", "person",
"audio"}
467
468
469     def test_cli_main_litmus_flag(tmp_path, monkeypatch, capsys):
470         """`--litmus` runs the static Gate-A config (single structural row)."""
471         d = str(tmp_path / "feat")
472         os.makedirs(d)
473         _write_learned_corpus(d)
474         monkeypatch.setattr("sys.argv", ["ablation", "--features-dir", d, "--litmus"])
475         ablation.main()
476         out = capsys.readouterr().out
477         assert "gate_a" in out
478         assert "static" in out                                     # litmus full set is the static
gate variant
479
480
481     def test_cli_main_needs_two_videos(tmp_path, monkeypatch):
482         d = str(tmp_path / "feat")
483         os.makedirs(d)
484         _write_learned_corpus(d, n=1)
485         monkeypatch.setattr("sys.argv", ["ablation", "--features-dir", d])
486         with pytest.raises(SystemExit):

```

26. assistant (2026-06-16T03:55:48.539Z)

Now I have everything I need. Let me confirm two final technical claims that the determinism section rests on: (1) whether the harness actually trains a `LogisticRegression` (a nondeterminism source the design completely ignores), and (2) whether floats round-trip. Let me check the experiment scorer and look for sklearn/numpy usage.

27. user (2026-06-16T03:55:51.784Z)

```

===scorer / LogisticRegression / numpy in experiment.py===
11: - `backend.eval.classifier` ? `LogisticRegression` (the pinnable JSON model)
49:from backend.eval.classifier import LogisticRegression
97: - ``classifier_model`` ? path to a frozen `LogisticRegression` JSON. When set,
213:def predict(variant: Variant, vc: VideoCandidates,
214:             model: Optional[LogisticRegression] = None,
220:             ``model`` (an already-fitted `LogisticRegression`) is supplied by `compare` per LOVO
233:             model = LogisticRegression.load(variant.classifier_model)
243:def _calibrate_and_tune(variant: Variant, model: LogisticRegression,
281:def _train(variant: Variant, videos: Sequence[VideoCandidates], iou: float) ->
LogisticRegression:
293:     return LogisticRegression().fit(X, y, variant.feature_names)
319:     pinned_model = (LogisticRegression.load(variant.classifier_model)
332:     model: Optional[LogisticRegression] = _train(variant, others, iou)
===classifier.py===
1: """A tiny, dependency-free logistic-regression classifier (numpy only).
4:no heavy ML stack ? the project already depends on numpy, and sklearn is not
18:import numpy as np
21:class LogisticRegression:
22:     def __init__(self, lr: float = 0.1, epochs: int = 1000, l2: float = 1e-3):
26:         self.w: Optional[np.ndarray] = None
27:         self.b: float = 0.0
28:         self.mean: Optional[np.ndarray] = None
29:         self.std: Optional[np.ndarray] = None

```

```

33:     def _sigmoid(z: np.ndarray) -> np.ndarray:
34:         return 1.0 / (1.0 + np.exp(-np.clip(z, -30, 30)))
35:     def _standardize(self, X: np.ndarray) -> np.ndarray:
36:     def fit(self, X, y, feature_names: Optional[List[str]] = None,
37:             class_weight: str = "balanced") -> "LogisticRegression":
38:         X = np.asarray(X, dtype=float)
39:         y = np.asarray(y, dtype=float)
40:         sw = np.where(y == 1, w_pos, w_neg)
41:         self.w = np.zeros(d)
42:         grad_b = float(err.mean())
43:     def predict_proba(self, X) -> np.ndarray:
44:         Xs = self._standardize(np.asarray(X, dtype=float))
45:     def predict(self, X, threshold: float = 0.5) -> np.ndarray:
46:     def from_dict(cls, d: dict) -> "LogisticRegression":
47:         m.w = np.asarray(d["w"], dtype=float) if d.get("w") is not None else None
48:         m.b = float(d.get("b", 0.0))
49:         m.mean = np.asarray(d["mean"], dtype=float) if d.get("mean") is not None else None
50:         m.std = np.asarray(d["std"], dtype=float) if d.get("std") is not None else None
51:     def load(cls, path: str) -> "LogisticRegression":

```

28. assistant (2026-06-16T03:55:58.773Z)

This is a decisive finding. The Tier-0 "unconditional" test trains a `LogisticRegression` via `\_train` ? `fit` using **numpy gradient descent over 1000 epochs** (`np.exp`, `np.clip`, matrix ops). This is a major hole in the determinism story: the design's serialization-determinism section quantizes the **fixture inputs** and **snapshot outputs**, but the **prediction path itself** runs platform-dependent **numpy BLAS** floating-point accumulation. A snapshot of **expected** segments computed on Linux can differ bit-for-bit from one computed on Windows/mac near a decision boundary. The design claims "byte-stable snapshot" but never addresses BLAS/numpy nondeterminism in the scorer it replays.

I now have enough verified evidence to write a relentless, grounded attack. Let me compile the structured findings.

29. user (2026-06-16T03:58:21.462Z)

Output does not match required schema: root: must have required property 'mustFixBeforeShipping'

30. assistant (2026-06-16T03:58:24.600Z)

The `mustFixBeforeShipping` field needs to be inside the same parameters object ? it appears my call split it. Let me resubmit with all properties in one parameters block.

31. user (2026-06-16T03:59:50.544Z)

Output does not match required schema: root: must have required property 'mustFixBeforeShipping'

32. assistant (2026-06-16T03:59:55.052Z)

The schema shows `attacks`, `overallVerdict`, and `mustFixBeforeShipping` as siblings, but the validator keeps reporting `mustFixBeforeShipping` missing ? likely the very long `overallVerdict` string is causing a parse truncation that drops the trailing property. Let me shorten the strings and ensure all three top-level properties are present and parse cleanly.

33. user (2026-06-16T04:00:59.832Z)

Output does not match required schema: root: must have required property 'mustFixBeforeShipping'

34. assistant (2026-06-16T04:01:03.962Z)

The validator keeps reporting `mustFixBeforeShipping` missing despite it being present. This is likely a parsing issue with the large payload ? possibly the `attacks` array (with its long strings and many entries) is causing truncation before the later siblings are parsed. Let me drastically shorten the payload: fewer/shorter attack entries and concise other fields, to

ensure all three top-level properties survive parsing.

35. user (2026-06-16T04:01:46.665Z)

Output does not match required schema: root: must have required property 'mustFixBeforeShipping'

36. assistant (2026-06-16T04:01:51.012Z)

The error persists even with a shorter payload, which suggests the issue is structural rather than size. Looking carefully: in my calls, `attacks` and `overallVerdict` use the `name=` attribute form while `mustFixBeforeShipping` uses `parameter name=`. The two parameters before it may be consuming the closing in a way that drops the third. Let me try ordering `mustFixBeforeShipping` first to confirm it parses when it isn't last.

37. user (2026-06-16T04:02:33.671Z)

Output does not match required schema: root: must have required property 'attacks'

38. assistant (2026-06-16T04:02:39.688Z)

Now the error moved to `attacks` (the last property), confirming the harness drops whatever property comes **last** in the call ? a serialization quirk truncating the final parameter. The fix is to keep all three properties but make the last-positioned one trivially small, then put the bulk earlier. Let me put `mustFixBeforeShipping` and `overallVerdict` first with full content, and `attacks` last but with compact entries to minimize truncation risk ? and critically, retry by reducing total size substantially so nothing is dropped.

39. user (2026-06-16T04:03:13.263Z)

Output does not match required schema: root: must have required property 'attacks', root: must have required property 'mustFixBeforeShipping'

40. assistant (2026-06-16T04:03:19.994Z)

Now `attacks` is missing (I omitted it to test) and `mustFixBeforeShipping` is reported missing again even though it's present and second. This confirms the harness is dropping the LAST property regardless of which it is. The robust workaround: make the final-positioned property tiny so its truncation doesn't lose content, and never let a content-heavy property sit last. But since the last property keeps getting dropped entirely, I need all three present with the last one being small. Let me order: full `overallVerdict`, full `mustFixBeforeShipping`, full `attacks` last ? but compress `attacks` heavily and put it last, accepting it must fully survive. Given the consistent drop-the-last behavior, let me instead try drastically smaller total payload (the real root cause is likely total size).

41. user (2026-06-16T04:03:45.024Z)

Structured output provided successfully

42. assistant (2026-06-16T04:04:01.300Z)

The structured output was accepted. Here is a summary of my red-team findings.

Verdict: Defensible foundation, not ship-ready as written

I verified every load-bearing code claim against the actual repo before attacking. The architecture's instincts are sound ? de-attribution-at-source doing the real work, encryption correctly scoped as a public-only barrier, the honest relative/conditional/revocable framing of non-attributability, and treating counsel/Fork-A/minors as preconditions it does not satisfy. But three load-bearing **technical** claims fail against the code:

HIGH severity (must fix):

1. **Determinism is false for the learned path.** The "unconditional, always-green, byte-stable cross-OS" Tier-0 test trains a numpy/BLAS `LogisticRegression`

(`backend/eval/classifier.py:18-81`, 1000 epochs of gradient descent). The design's quantization only touches fixture inputs and serialized bytes ? never the float accumulation inside `fit()`. Near `threshold=0.5`, BLAS order/thread differences flip `predict()` between the Windows GPU box and the ubuntu-only CI, so a blessed snapshot can fail on Linux. This is the exact "green proves no side effects" trap the brief targets.

2. ****Leak remediation is too narrow.**** The design scrubs only

`eval\_baselines/heuristic\_lovo\_n6.json`, but the same real player names are committed in `eval\_baselines/gemini\_wrapper\_2026-06-10.json`, `backend/eval/eval\_partial\_labels.py:23,119` (line 23 also leaks a full `C:/Users/avidu/...` local path), and `tests/test\_cleanup\_caches.py` ? and in git history since PR #77 (commit `d51efe3`).

3. ****Pre-commit hook is mis-cast as a barrier.**** No hook infra exists (`.pre-commit-config.yaml` / `.github/hooks/` absent); `--no-verify` and fresh clones bypass it. Only a ****required**** CI status check is enforceable, and the design never specifies one.

****MEDIUM (privacy blind spots):**** the Tier-0 loaders (`fusion\_compare.py:47-48`, `ablation.py:400-401`) read `c['person']` unconditionally ? contradicting "strip person, reuse unchanged"; `experiment.\_feature\_column:180-192` silently defaults a missing column to `0.0` so green cannot **prove** sensitive columns were stripped; the HMAC salt is brute-forceable over the tiny convention-named roster the instant it leaks (use a random `os.urandom` surrogate instead); and retained `fps`/`frame\_width` plus the per-fixture duration/crossing/GT vector are a camera/venue/match quasi-identifier at n=6.

Key files referenced: `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\classifier.py`, `backend\eval\fusion\_compare.py`, `backend\eval\ablation.py`, `backend\eval\experiment.py`, `backend\eval\metrics.py`, `eval\_baselines\heuristic\_lovo\_n6.json`, `eval\_baselines\gemini\_wrapper\_2026-06-10.json`, `backend\eval\eval\_partial\_labels.py`, `.github\workflows\ci.yml`.

The legal gating (counsel sign-off, Fork-A-only, minors/biometric exclusion) is correctly treated as a precondition the design does not satisfy ? that posture should stay intact, and an engineering-green PR must never be read as legal clearance.